

좋은 기계란 ? (Good Machine)

High quality

- High performance.
- multi - function.
- High efficiency
- Low price

Easy to make

parts, material, machine tool,
assembly, cost,

Easy to use

↳ operation, tuning, inspection, repair, cleaning.
less breakdown/malfunction, noise, size

Easy to throw away / dispose of capability of recycling.

environmentally friendly
sustainability.

Machine : A device to transform / transfer physical quantities
and to be useful for people.

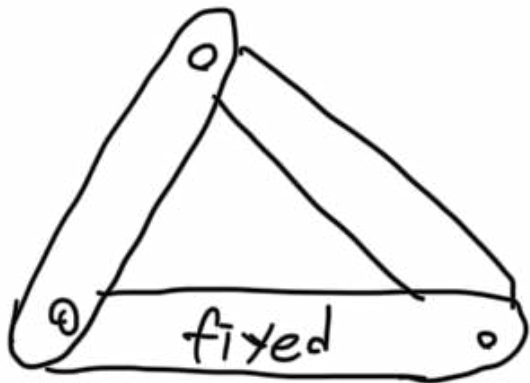
Input $\rightarrow$ Mechanism $\rightarrow$ Output.

Mechanism: device that has the purpose of transferring motion / force from the input to output.

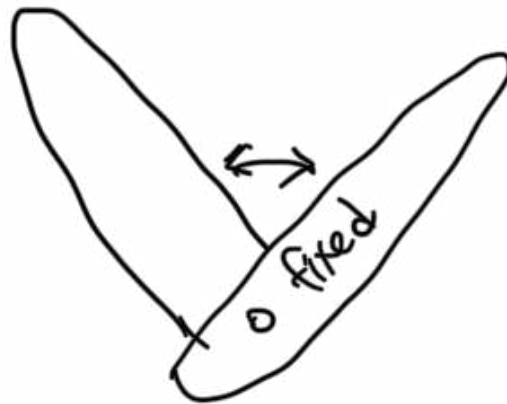


Linkage: Σ link, bar, rigid connected by joints to form open/closed chains/loops.

Kinematic chains, with at least one link fixed, become mechanisms or structures depending on mobility ($\# \text{ of mobile links} \geq 2$).

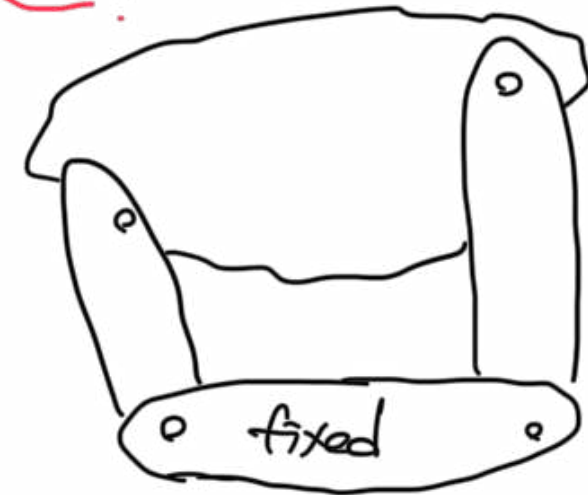


No mobility $\Rightarrow$ Structure



No mobility (number of mobile links < 2)

$\Rightarrow$ 뚫라 아님! (정비에 재가 재작도?)



$\Rightarrow$ mechanism, (closed chain)

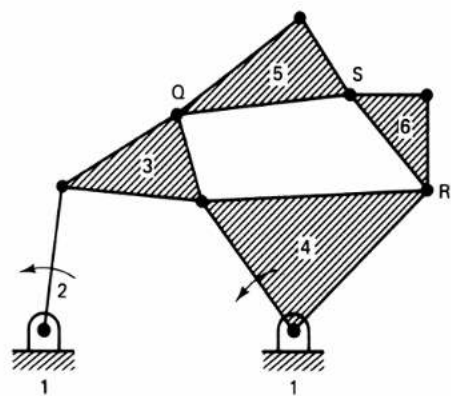
Machine = Σ Mechanism.

* 고정 link 에 따라 link 사이의 상대운동은 불변

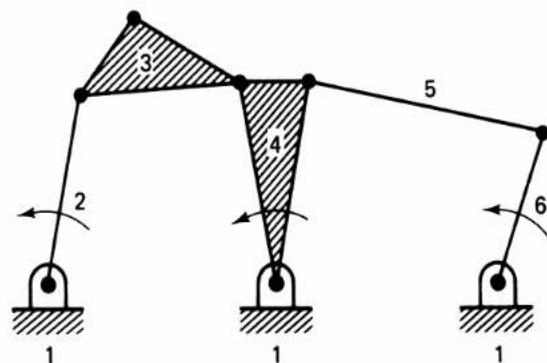
< Plane Motion > : 2D plate 위에서만 운동. = translation + rotation.

Lower pairs : full 접촉 joint : Surface Contact pairs.

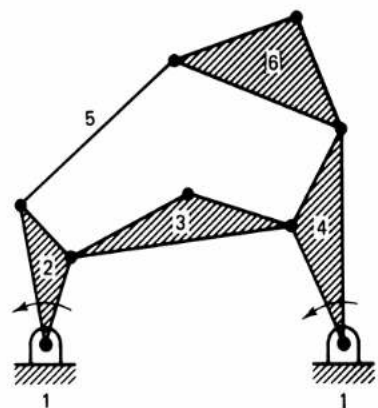
Higher Pairs (joints) : have line/point contact. E.g) cam, ball, roller bearings, gear



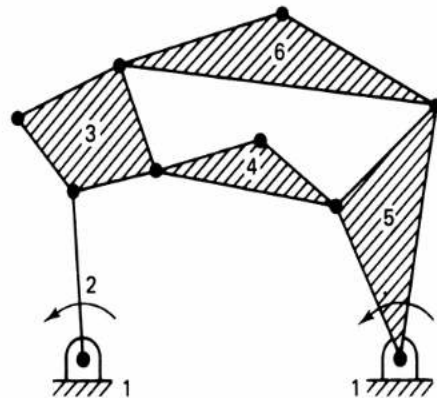
Watt I six-bar linkage.



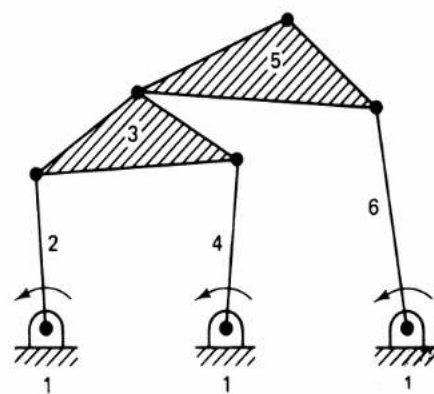
Watt II six-bar linkage



Stephenson I six-bar linkage



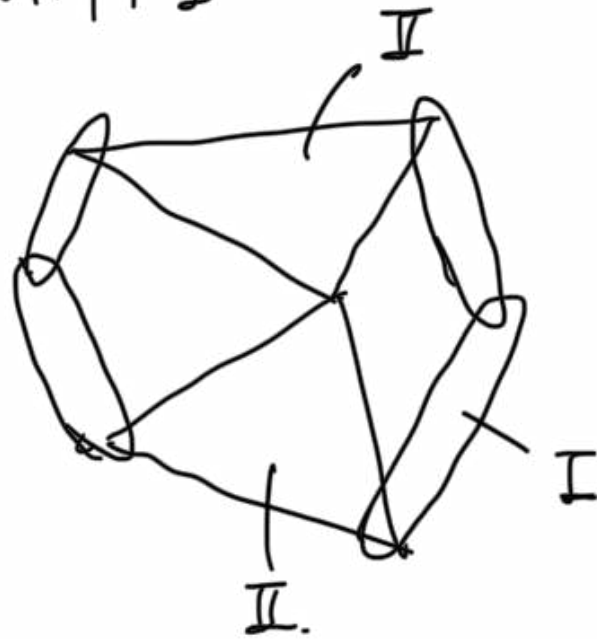
Stephenson II six-bar linkage



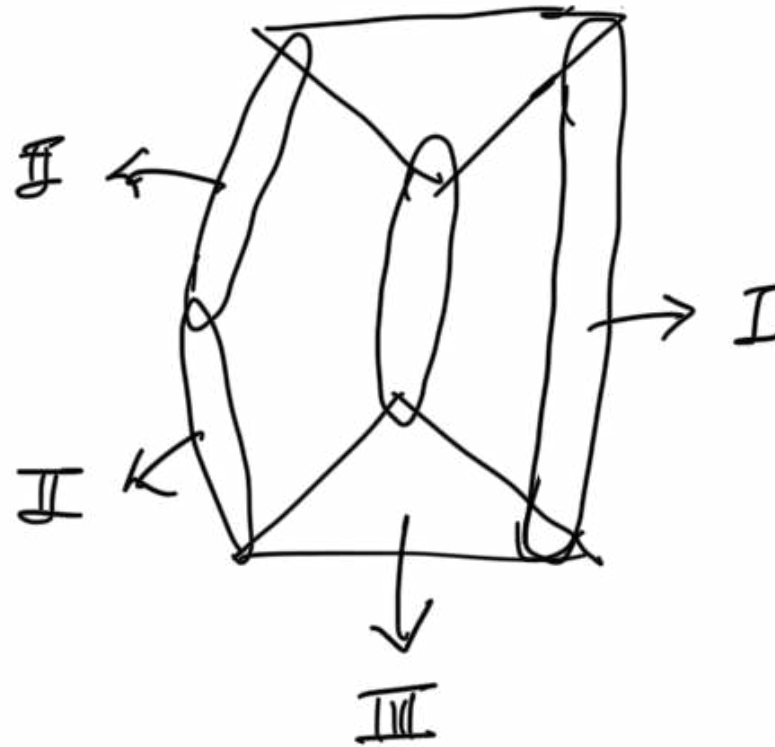
Watt : Ternary link 서로 직결.

고정에 따라 $I \sim II$ 이 결정.

WATT I

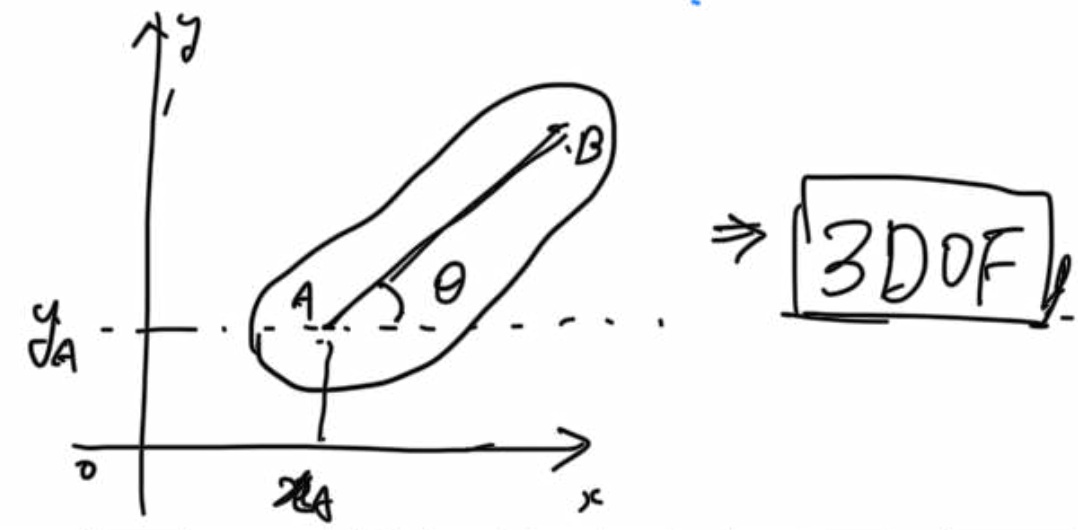


STEPHENSON.

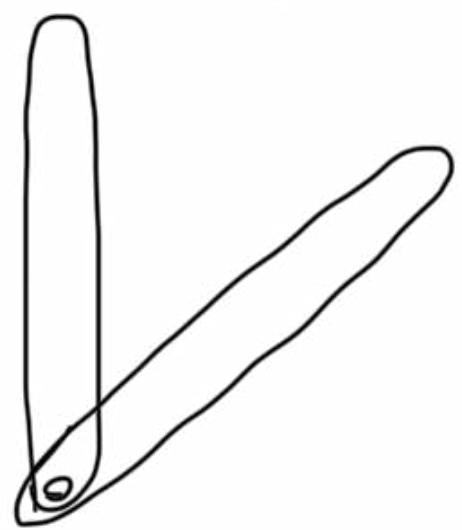


Degrees of Freedom

: Number of independent inputs required to determine the position of all links of the mechanism with respect to ground.



Gruebler's Equation



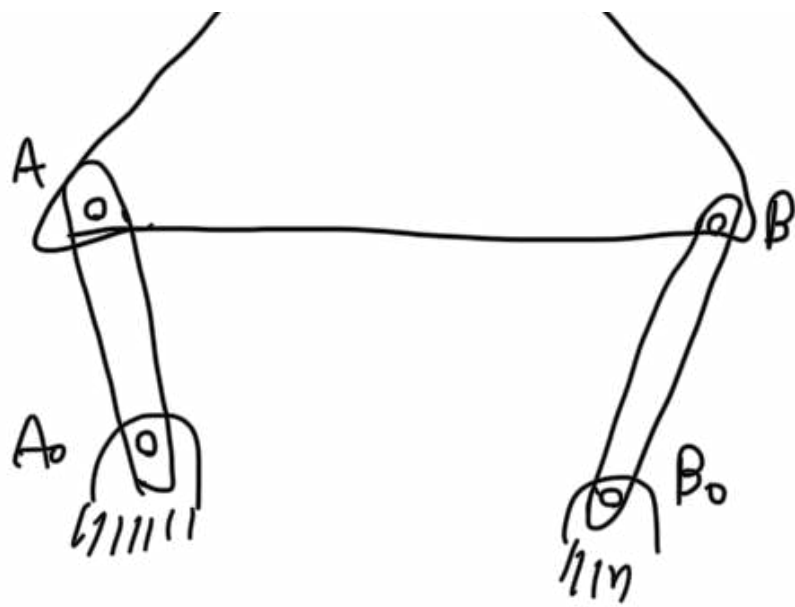
n links, f pin joints
 one link fixed on ground

$$DOF : F = 3(n-1) - 2f \quad (\text{Gruebler's Eq})$$

$4 DOF = 3 - 2 - 2 \cdot 1 = 4$ * pin joint = lower link

E.g.)

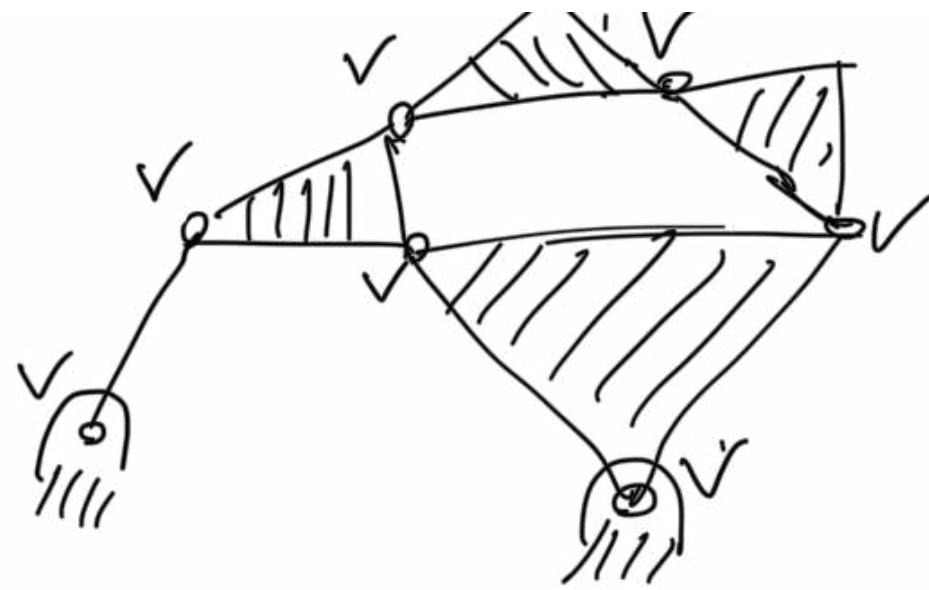




$$F = 3 \cdot 3 - 2 \cdot 4 = 1 \quad (\text{Gruebler})$$

A fixed $\rightarrow$ If A fixed $\rightarrow$ B should be fixed

$\Rightarrow$ 1 DOF (Intuition)



$$F = 3 \cdot 5 - 2 \cdot 7 = 1 \quad (\text{Gruebler})$$

* Also can be done intuitively.

Has to be $F = 3$.

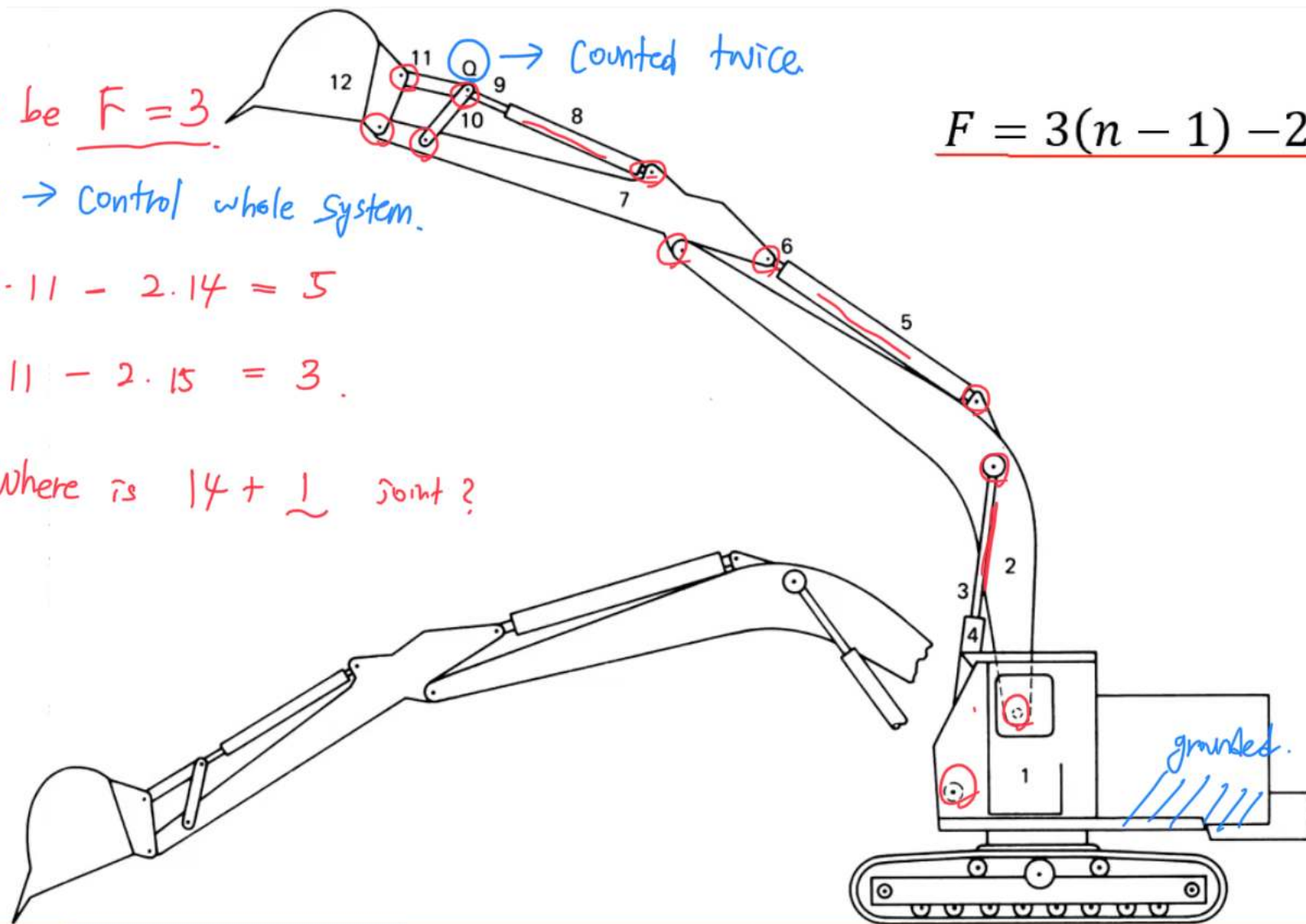
$\therefore 3, 5, 8 \rightarrow$ Control whole system.

$$3 \cdot 11 - 2 \cdot 14 = 5$$

$$3 \cdot 11 - 2 \cdot 15 = 3$$

Q) Where is $14 + 1$ joint?

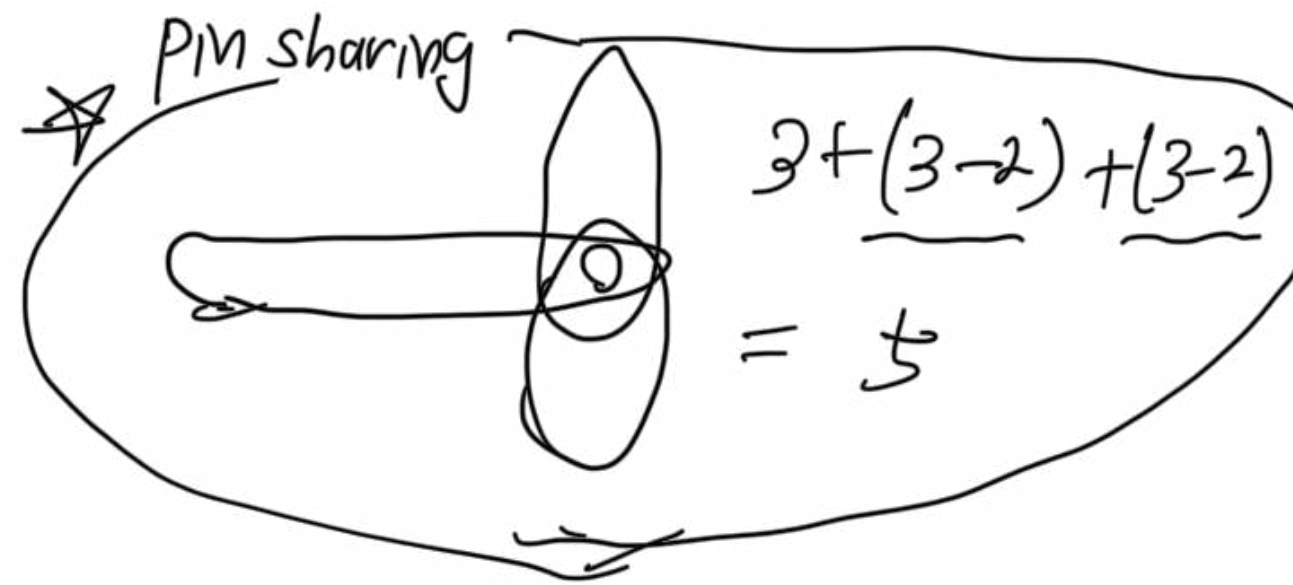
$$\underline{F = 3(n - 1) - 2f_1}$$



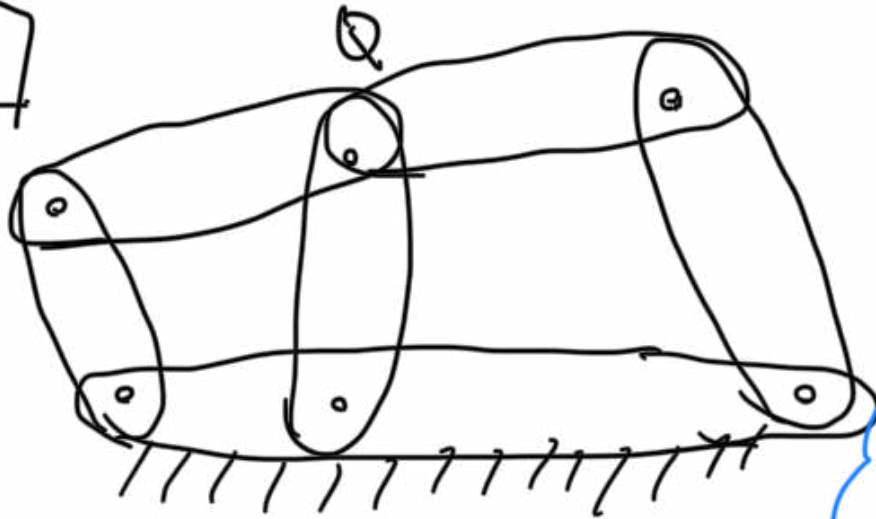
① $7 \sim 12 \Rightarrow 3 \cdot 5 - 2 \cdot 7$ ("1" fixed)

② $2, 5, 6, 7 \Rightarrow 3 \cdot 3 - 2 \cdot 4 = 1$ ("2" fixed)

③ $1 \sim 4 \Rightarrow 3 \cdot 3 - 2 \cdot 4 = 1$ ("1" fixed)



7 ~ 12



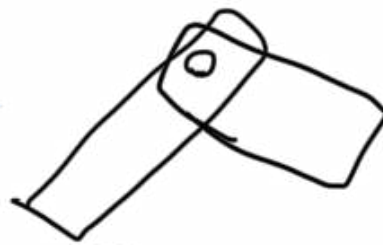
2 link $\Rightarrow$ 1 pin joint.



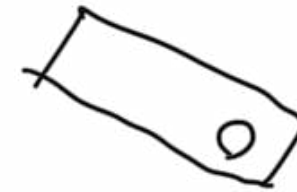
pin joint shared by links more than two should be thought twice.

$F = 3 \cdot 5 - 2 \cdot 6$ (X)

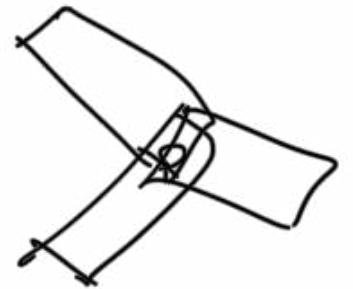
$F = 3 \cdot 5 - 2 \cdot 7$ (O)



+



=

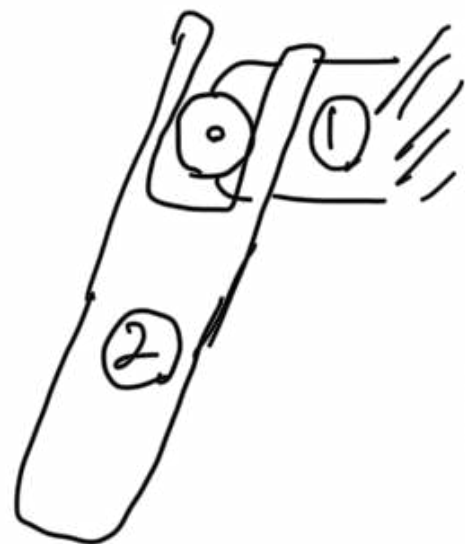


$\therefore$ Gruebler's Equation : $F = \underbrace{3(n-1)}_{\text{links}} - \underbrace{2f_1}_{\text{lower pair}} - \underbrace{f_2}_{\text{higher pair}}$

(modified)

pin joint & slide = f_1

- 0'



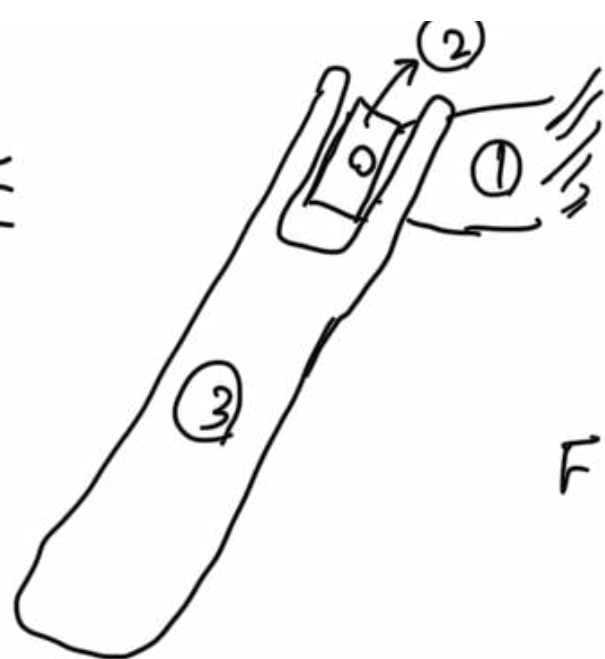
$$n = 2$$

$$f_1 = 0$$

$$f_2 = 1$$

$$F = 3(2-1) - 1 = \underline{2}$$

$\equiv$



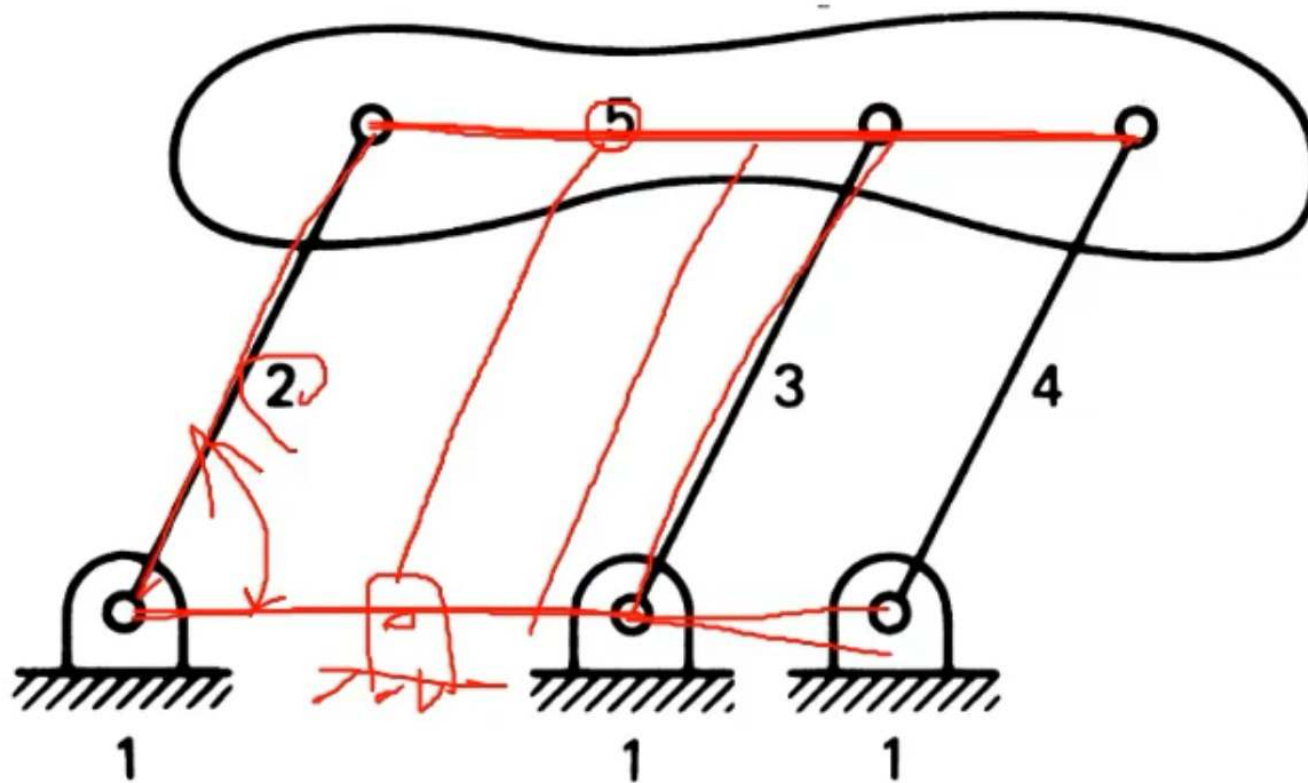
$$n = 3$$

$$f_1 = 2$$

$$f_2 = 0$$

$$F = 3(3-1) - 2 \cdot 2 = \underline{2}$$

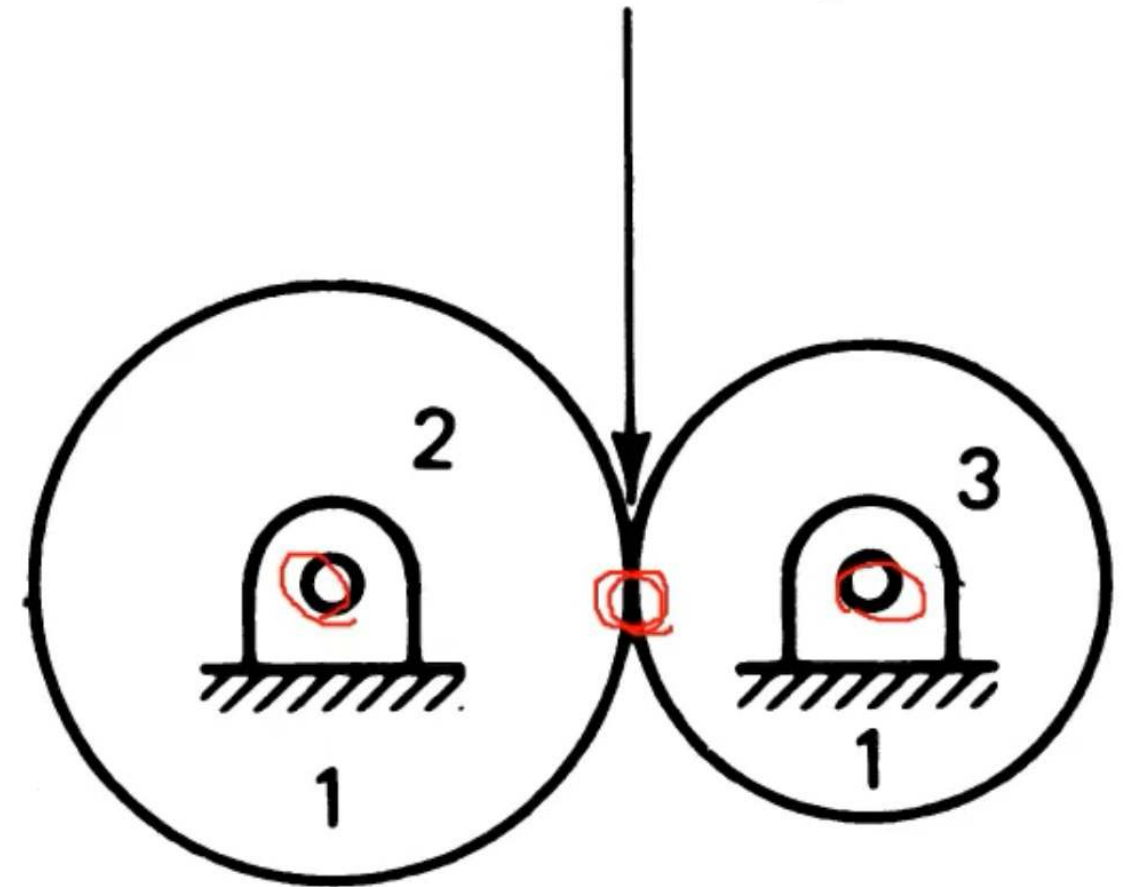
Overconstrained linkage



$$F = 3 \times (5 - 1) - 2 \times 6 = 0$$

$F = 1$ when all parallel to each other.

Pure Rolling



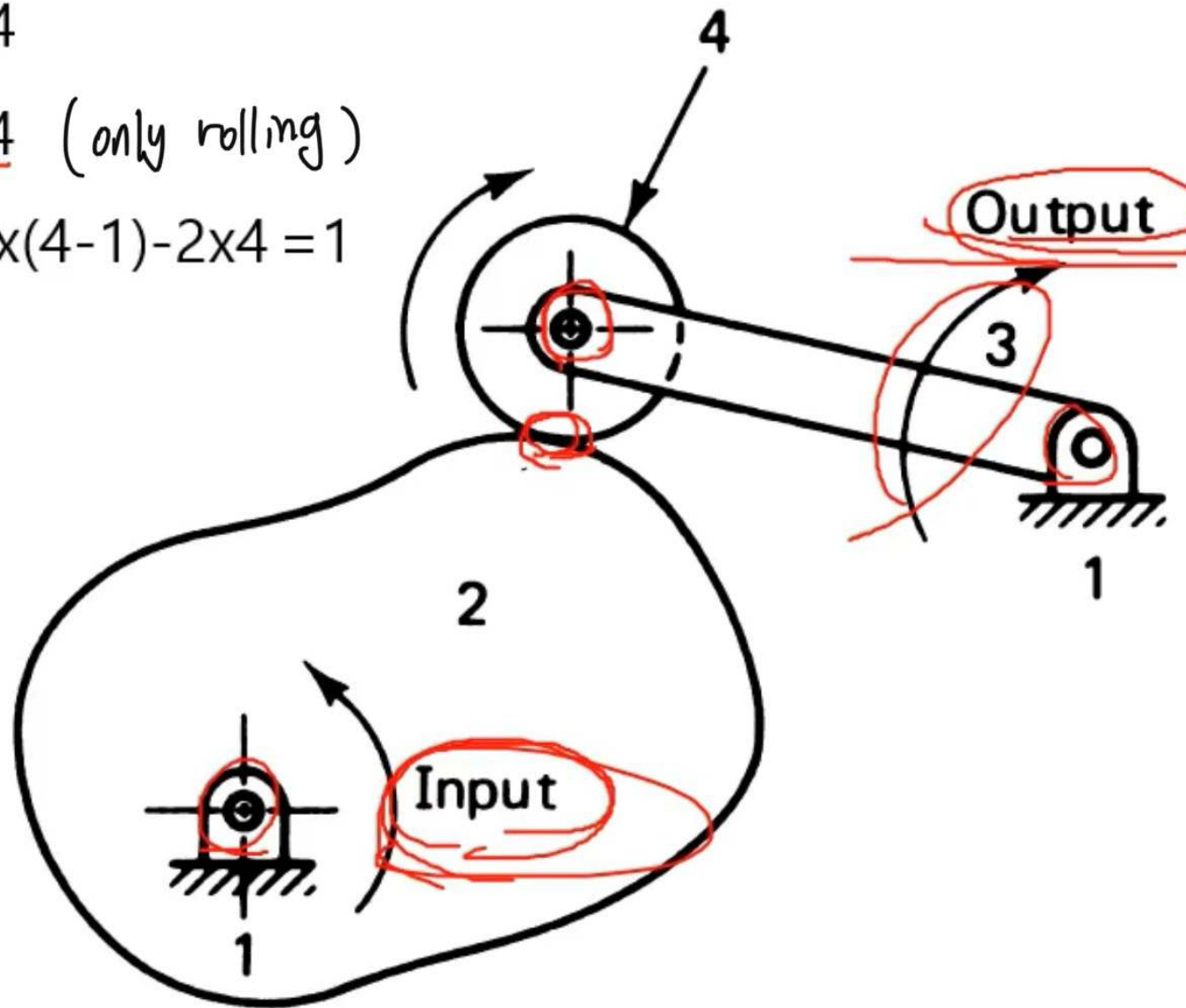
$$F = 3 \times (3 - 1) - 2 \times 3 = 0$$

Passive or redundant degree of freedom

$$n = 4$$

$$f_1 = 4 \text{ (only rolling)}$$

$$F = 3 \times (4 - 1) - 2 \times 4 = 1$$



$$n = 4$$

$$f_1 = 3$$

$$f_2 = 1 \text{ (allows sliding + rolling)}$$

$$F = 3 \times (4 - 1) - 2 \times 3 - 1 = \underline{2}$$

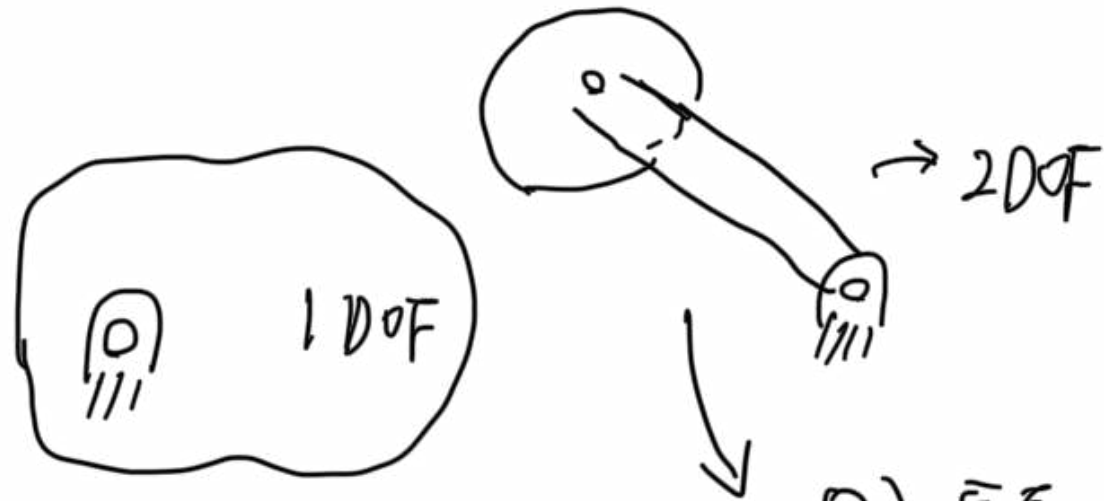
$$n = 3 \text{ (③ \& ④ fixed together)}$$

$$f_1 = 2$$

$$f_2 = 1 \text{ (only sliding)}$$

$$F = 3 \times (3 - 1) - 2 \times 2 - 1 = 1$$

(Q) what is DOF for whole system ?



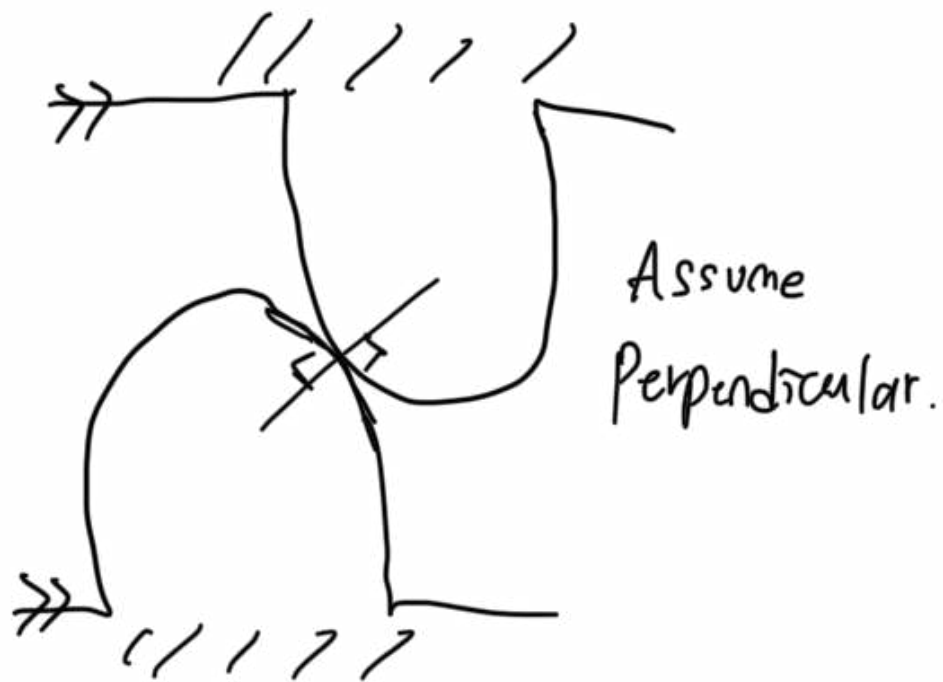
$$\left[\begin{array}{l} \text{비접촉} : n=4, f_1=3 \Rightarrow \underline{F=3} \\ \text{접촉 (no slide)} : n=4, f_1=4 \Rightarrow \underline{F=1} \end{array} \right]$$

(Q) 동동 두면 어떻게 할?

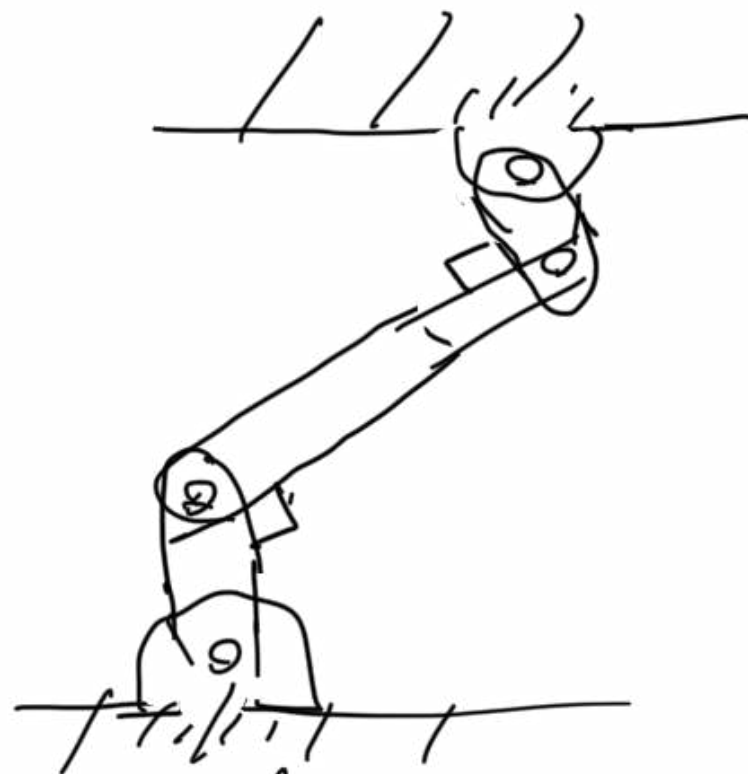
DOF determination & kinematic diagram drawing are

First Step in both kinematic analysis & synthesis processes.

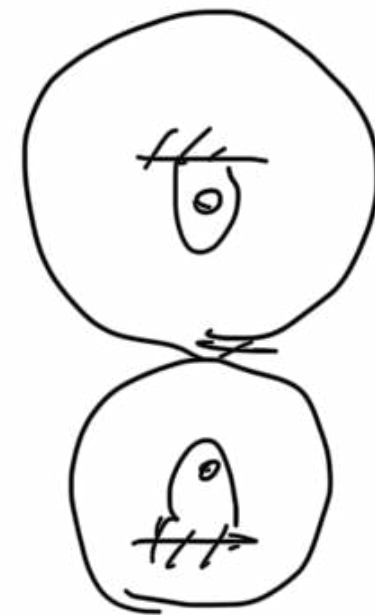
Higher Pair : DOF 1개 줄어 / Lower Pair : DOF 2개 줄어.



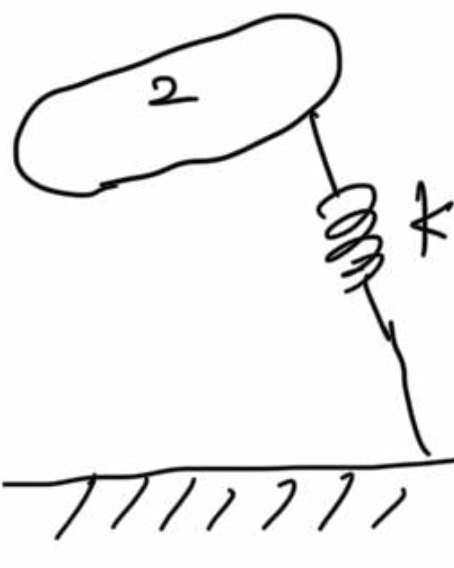
$$3 \cdot (2 - 1) - 1 = 2$$



$$3 \cdot (4 - 1) - 2 \cdot 4 = 1$$



$$3 \cdot (3 - 1) - 2 \cdot 2 - 1 \cdot 1 = 1$$

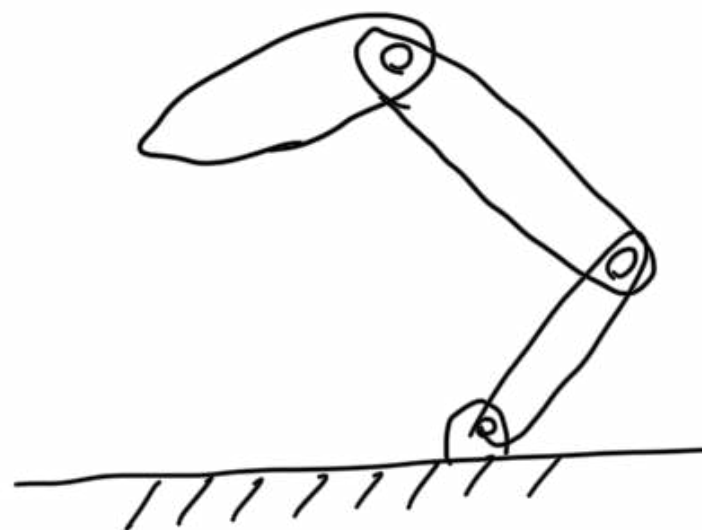


$$n = 2$$

$$f_1 = 0$$

$$f_2 = 0$$

$$F = 3$$



$$n = 4$$

$$f_1 = 3$$

$$f_2 = 0$$

$$F = 3$$

* spring does not affect # of DOFs (only force exertion).

① Why 4-bar linkage is simplest closed-loop.

$$F = 3(n-1) - 2f_1 - f_2, \quad f_2 = 0 \text{ assuming,}$$

$$\Rightarrow F = 3(n-1) - 2f_1, \text{ for } F=1 \text{ purpose, } 3n = \underbrace{2f_1 + 4}_{\text{even}} \Rightarrow n = \text{even \#}$$

To be closed loop, $n \geq 3 \Rightarrow n=4 \Rightarrow$ 4 bar linkage is simplest closed loop.

② How to figure out linkage components

$$n = B + T + Q + p + H + \dots$$

$$2f_1 = 2B + 3T + 4Q + \dots$$

B : binary
T : Tertiary
⋮

Using $F = 3(n-1) - 2f_1, \Rightarrow n - (F+3) = T + 2Q + 3p + 4H + \dots$

Thus, if $F=1$, and $n=4$, (4-bar linkage), only binary link possible.

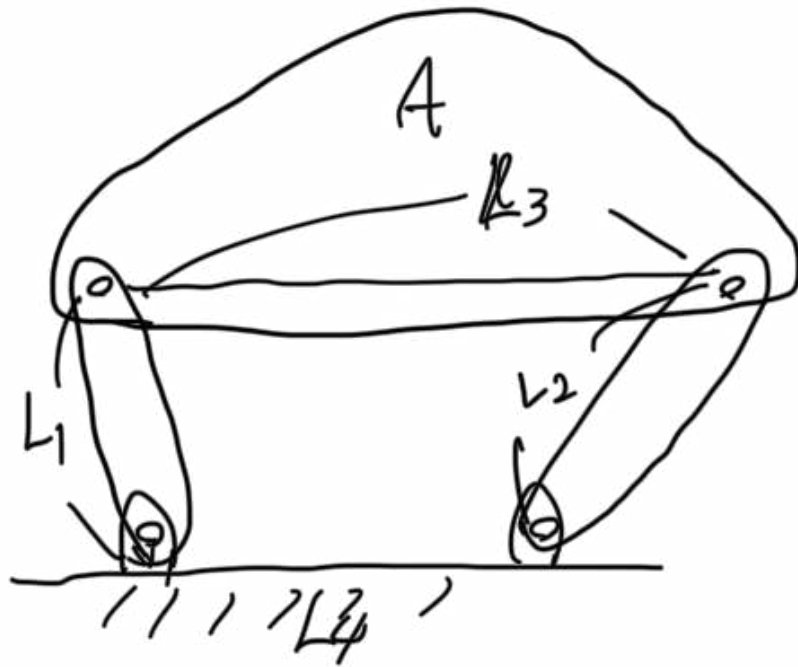
if $F=1$ and $n=6$, (6 "), $T \geq 2, B=4$

$\alpha = 1$, $B = 5 \Rightarrow$ 불가능!

HW1 $\star$ 1-2, 16, 36.

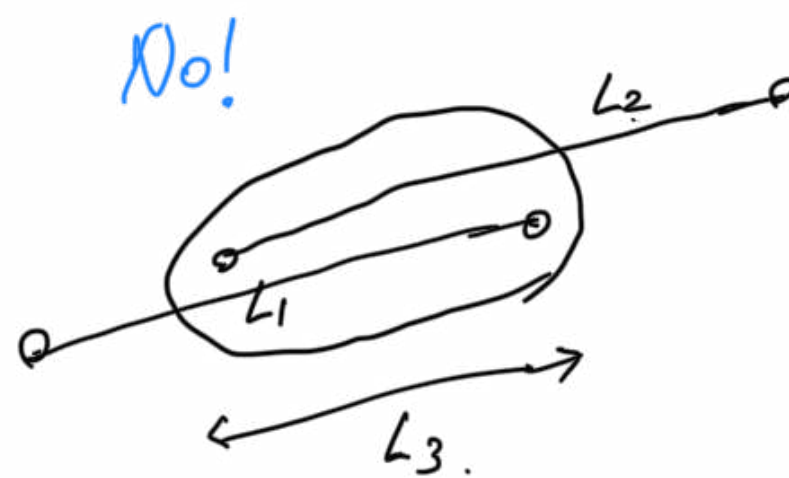
----- Chapter 1 -----

Q) Can "A" rotate?



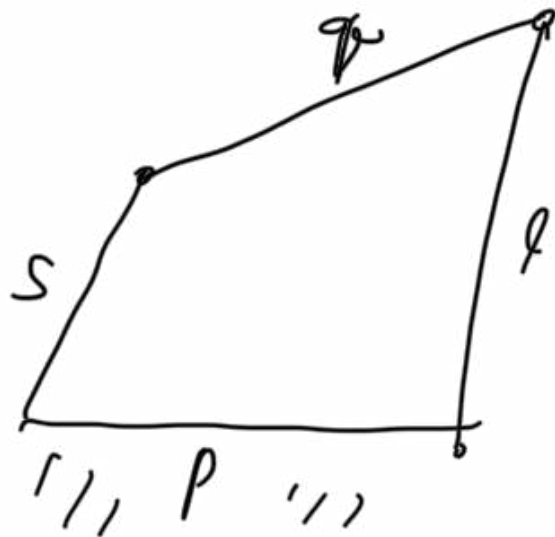
My idea : If $|L_1 - L_2| \equiv L_3 \rightarrow \text{rotate!}$

($\therefore$) **No!**



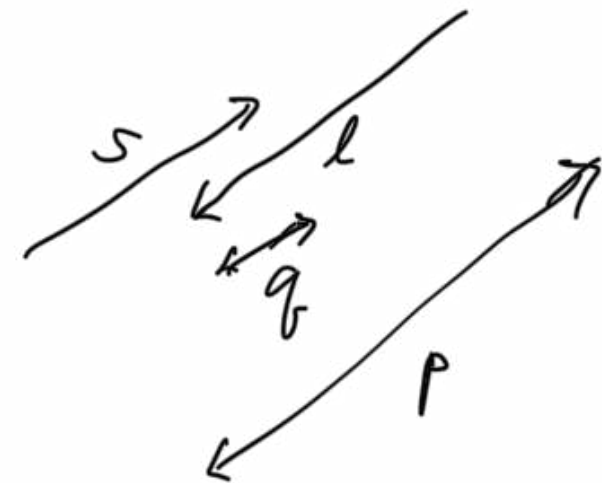
$L_1 < L_3, L_4 < L_2$?

: Boundary Condition.



Grashof criteria : $s + l < p + q$

$$\underline{s + l - q < p} \quad \text{pf)}$$

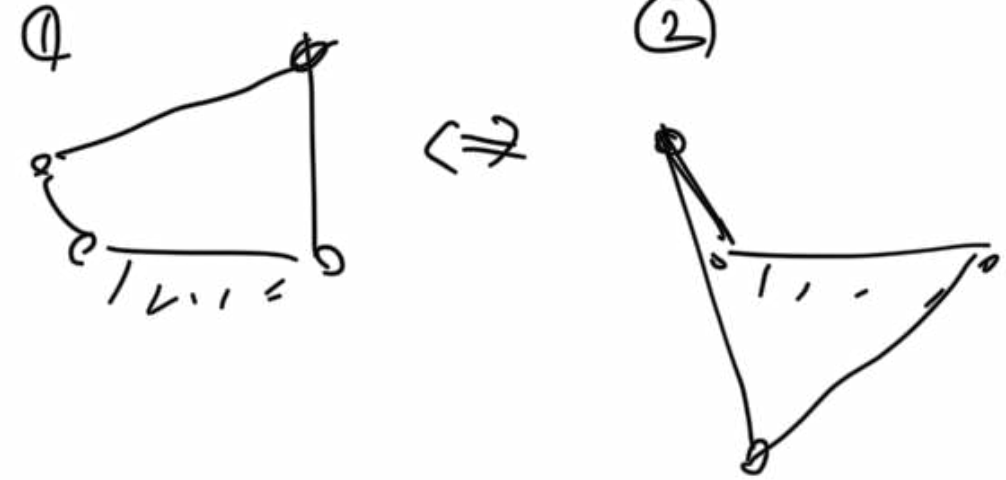


$\Rightarrow$ Kinematic inversion

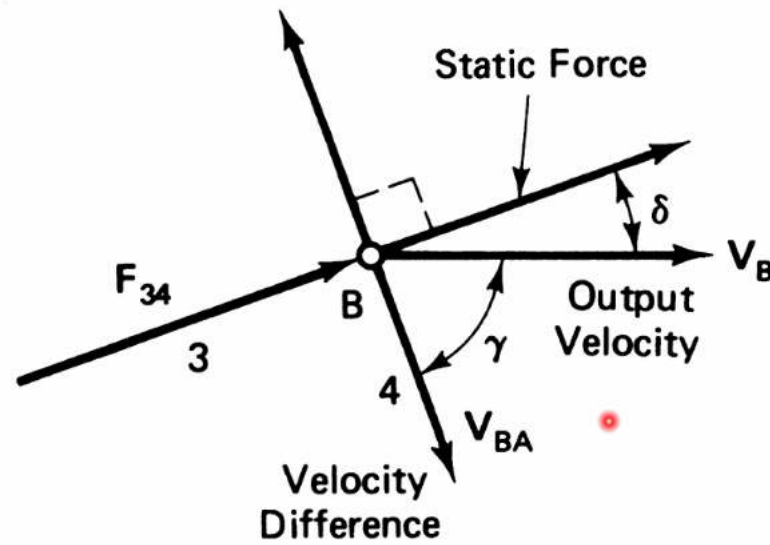
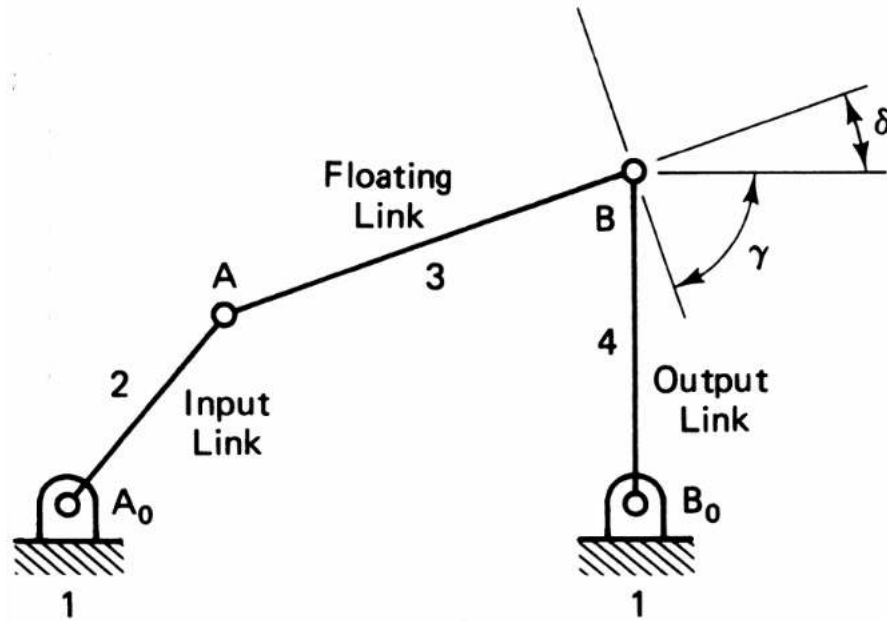
Geometric inversion.

고정 link 바깥

특정 linkage 에 대해
여러 가능성

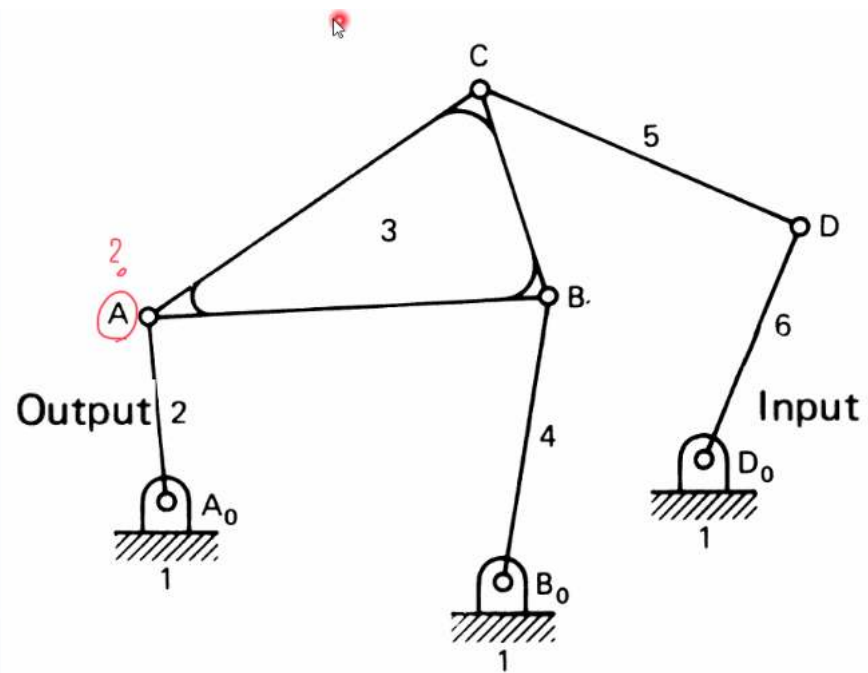


Transmission and deviation angles, γ and δ



//
A 에 대한 B의 상대속도 방향

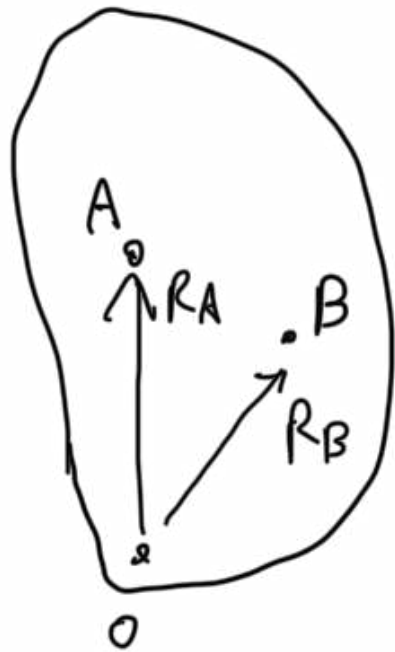
θ has to be close to 0 for force transfer



How ? (Input $\rightarrow$ output)

Same point		Different point
Same link 같은 링크에 있는.	Case 1 Trivial	Case 2 Difference motion
Different links 다른 링크에 있는	Case 3 Relative motion	Case 4 Manageable through a series of case 2 and case 3 steps

Link 는 얼마든지 확장 가능. ↗



$$\vec{V}_A = \frac{d}{dt} \{ \vec{R}_A \} = \frac{d}{dt} \{ R_A e^{j\omega t} \} = j\omega \vec{R}_A$$

$$\vec{V}_{BA} = \vec{V}_B - \vec{V}_A = j\omega (\vec{R}_B - \vec{R}_A) = j\omega \vec{R}_{BA}$$

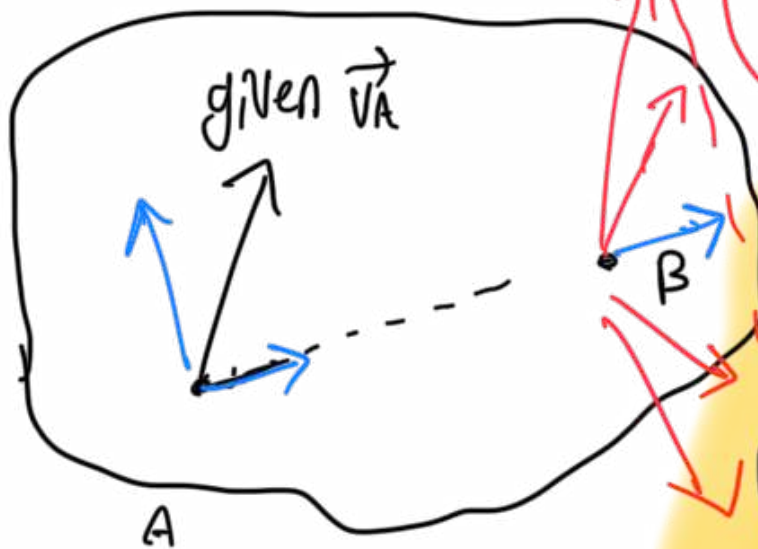
$$\Rightarrow \omega = \frac{\vec{V}_{BA}}{j \vec{R}_{BA}} \Rightarrow |\omega| = \frac{\|\vec{V}_{BA}\|}{\|\vec{R}_{BA}\|}$$

Note : $\omega > 0$
= counter
clock wise

 $\omega > 0$ $\omega < 0$

$R_{BA}(A \text{ 가 기준!})$

B의 속도는
→ 만가능!



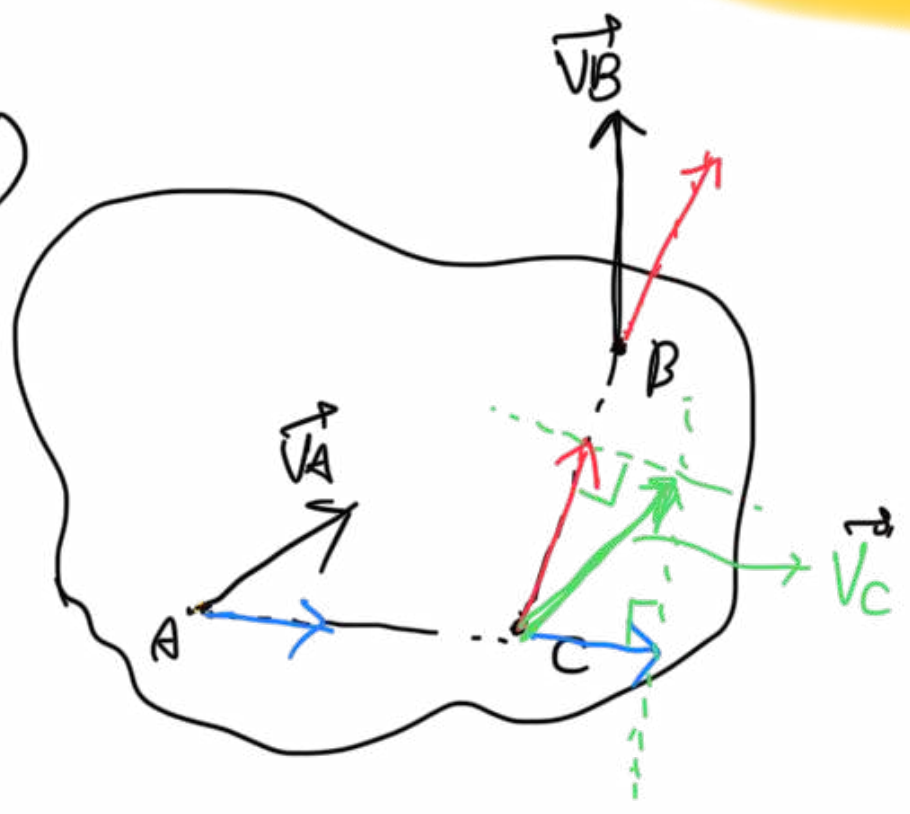
★ Note : AB 위 방향 속도 성분이 동일해야 함 ($\because$ Rigid Body).

why? $\rightarrow \omega = \frac{\vec{V}_{BA}}{j \vec{R}_{BA}} \in \mathbb{R}$

Thus $\vec{V}_B$ and $\vec{V}_A$ have to be satisfied

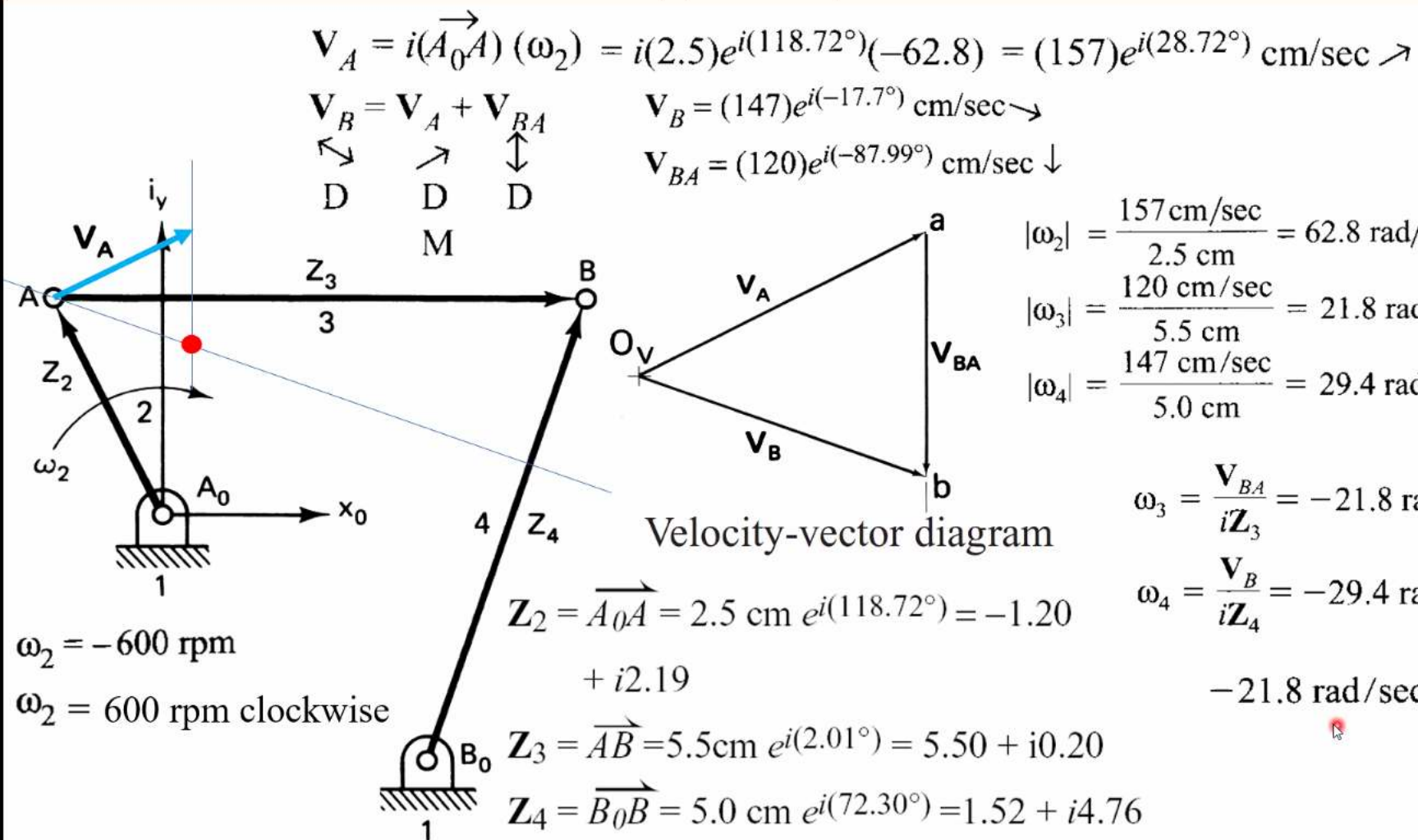
$\vec{v}_B, \vec{v}_{BA} \perp \vec{r}_{BA}$ has to be satisfied.

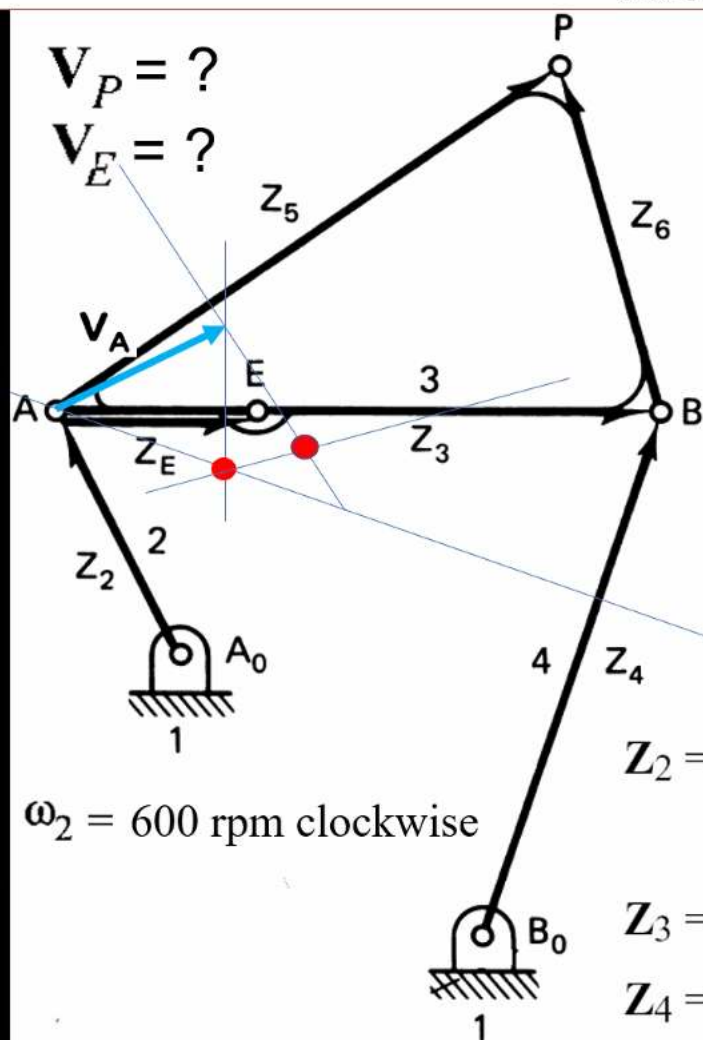
E.g.)



Note: In textbook, $\omega_2 = -600 \text{ rpm}$: clockwise. $\approx \left\{ \begin{array}{l} \curvearrowright \omega_2 \\ \omega_2 = \leftarrow 600 \text{ rpm} \end{array} \right\}$

$\curvearrowright$: arrow indicates purely direction!

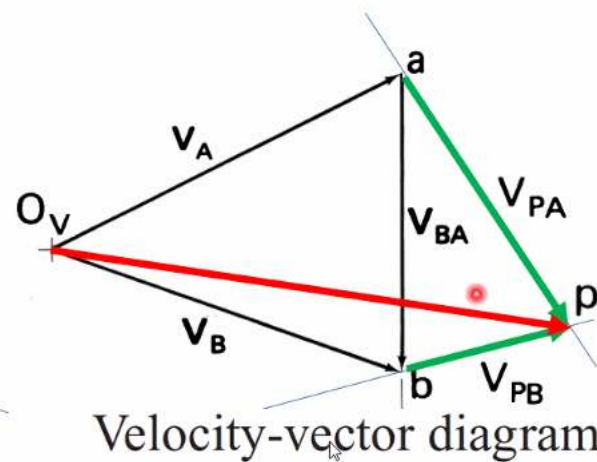




$$\mathbf{V}_P = \mathbf{V}_A + \mathbf{V}_{PA}$$

$$\mathbf{V}_B = \mathbf{V}_A + \mathbf{V}_{BA}$$

$$\mathbf{V}_P = \mathbf{V}_B + \mathbf{V}_{PB}$$



$$\mathbf{V}_A + \mathbf{V}_{PA} = \mathbf{V}_B + \mathbf{V}_{PB}$$

$$\mathbf{Z}_2 = \overrightarrow{A_0A} = 2.5 \text{ cm } e^{i(118.72^\circ)} = -1.20 + i2.19$$

$$\mathbf{Z}_3 = \overrightarrow{AB} = 5.5 \text{ cm } e^{i(2.01^\circ)} = 5.50 + i0.20$$

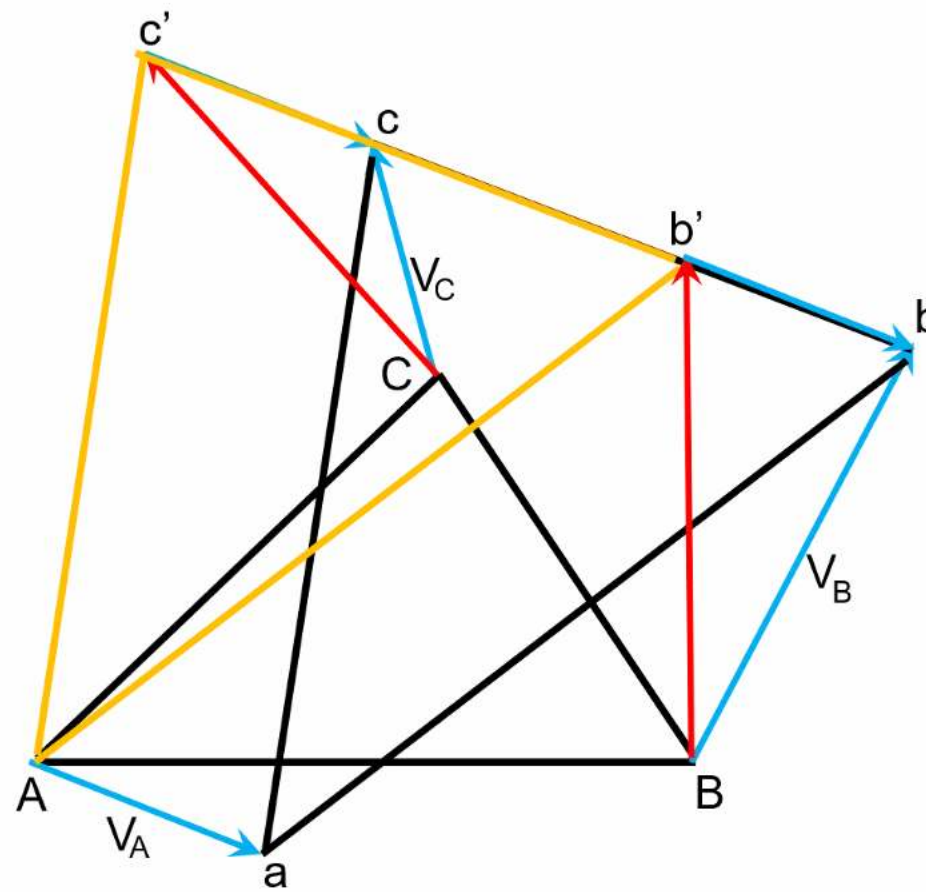
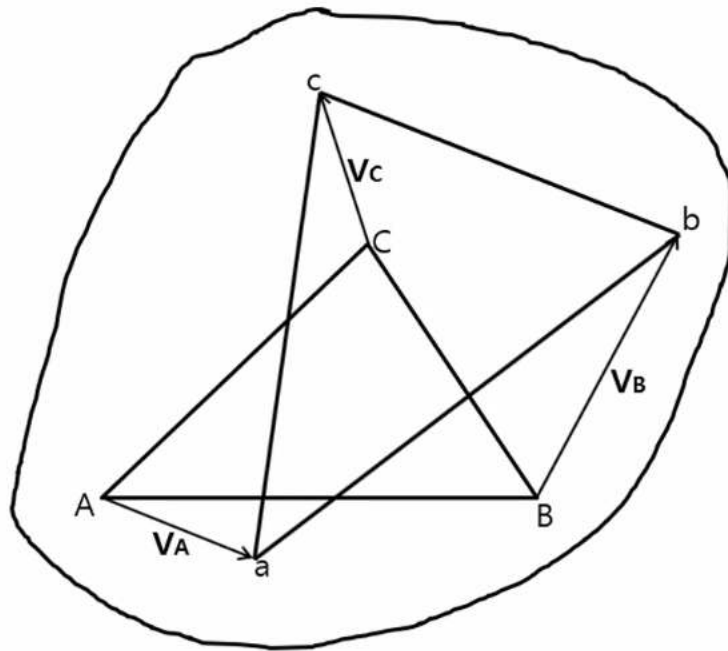
$$\mathbf{Z}_4 = \overrightarrow{B_0B} = 5.0 \text{ cm } e^{i(72.30^\circ)} = 1.52 + i4.76$$

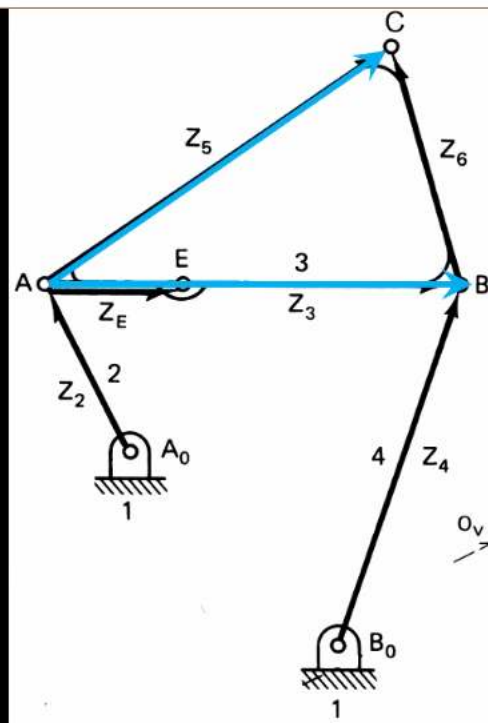
$$\triangle ABC \sim \triangle abc$$

$$\triangle Ab'c' = \triangle abc$$

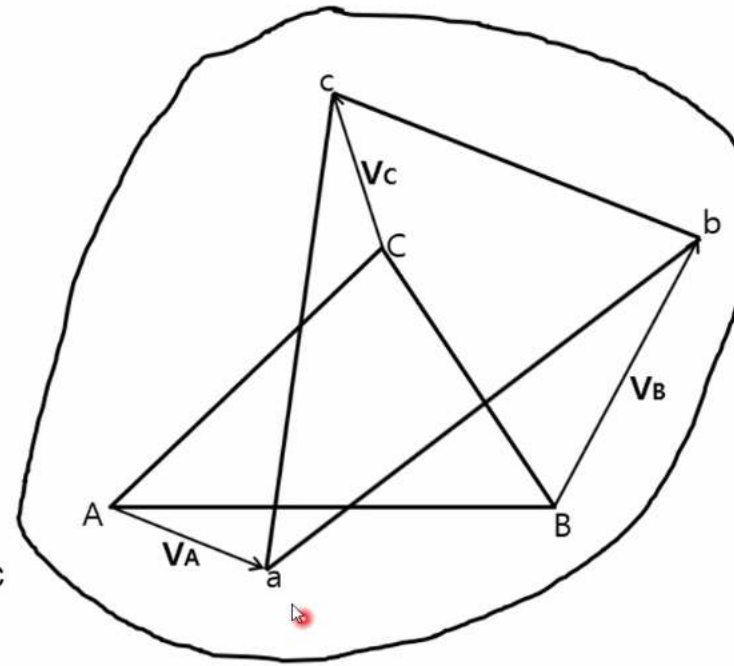
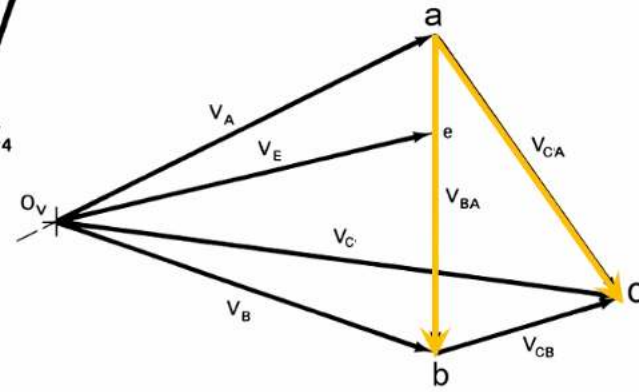
$$\triangle ABb' \sim \triangle ACc'$$

$$\triangle ABC \sim \triangle Ab'c'$$





$$\triangle ABC \sim \triangle abc$$



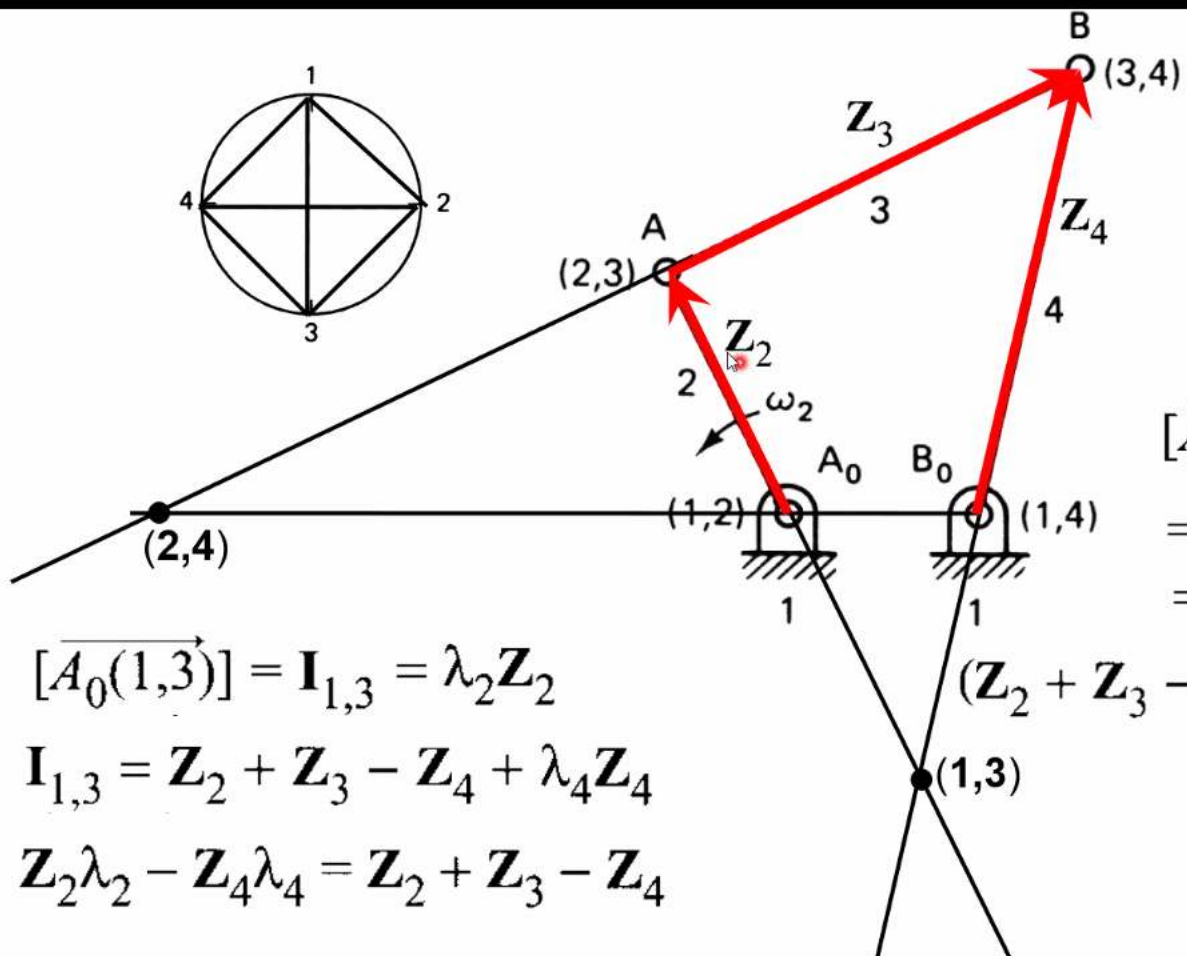
$$Z_C = Z_A + C(Z_B - Z_A) \quad \dot{Z}_C = \dot{Z}_A + C(\dot{Z}_B - \dot{Z}_A)$$

$$C = \frac{Z_C - Z_A}{Z_B - Z_A} = \frac{\dot{Z}_C - \dot{Z}_A}{\dot{Z}_B - \dot{Z}_A} = \frac{Z_C - Z_A + k(\dot{Z}_C - \dot{Z}_A)}{Z_B - Z_A + k(\dot{Z}_B - \dot{Z}_A)} = \frac{Z_C + k\dot{Z}_C - (Z_A + k\dot{Z}_A)}{Z_B + k\dot{Z}_B - (Z_A + k\dot{Z}_A)}$$

$$\frac{\omega_2}{\omega_3} = \frac{\overrightarrow{(O_3 Q)}}{\overrightarrow{(O_2 Q)}} = \frac{\overrightarrow{(1,3 - 2,3)}}{\overrightarrow{(1,2 - 2,3)}}$$

→ Simple proof.

Mechanism Design: Analysis and Synthesis, Volume I, 4th Ed.



$$[\overrightarrow{A_0(1,3)}] = \mathbf{I}_{1,3} = \lambda_2 \mathbf{Z}_2$$

$$\mathbf{I}_{1,3} = \mathbf{Z}_2 + \mathbf{Z}_3 - \mathbf{Z}_4 + \lambda_4 \mathbf{Z}_4$$

$$\mathbf{Z}_2 \lambda_2 - \mathbf{Z}_4 \lambda_4 = \mathbf{Z}_2 + \mathbf{Z}_3 - \mathbf{Z}_4$$

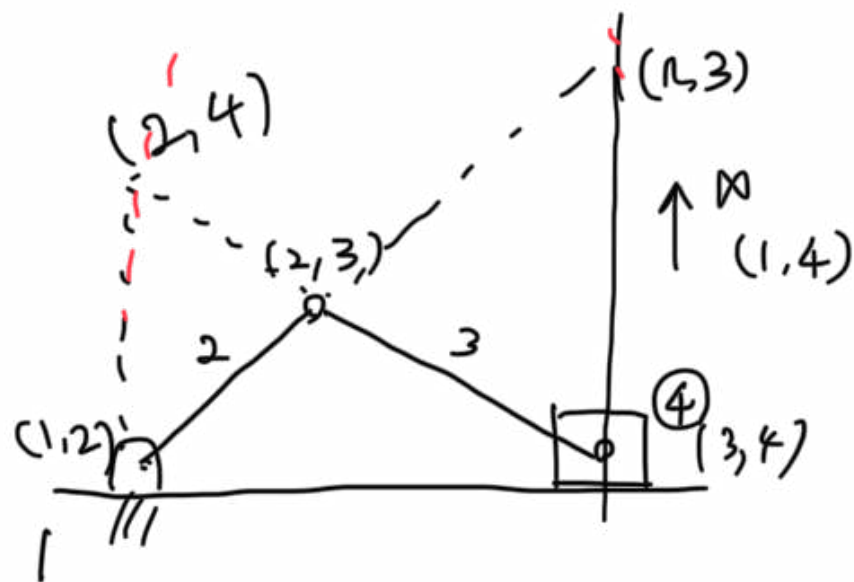
$$[\overrightarrow{A_0(2,4)}] = \mathbf{I}_{2,4}$$

$$= \lambda_1 (\mathbf{Z}_2 + \mathbf{Z}_3 - \mathbf{Z}_4)$$

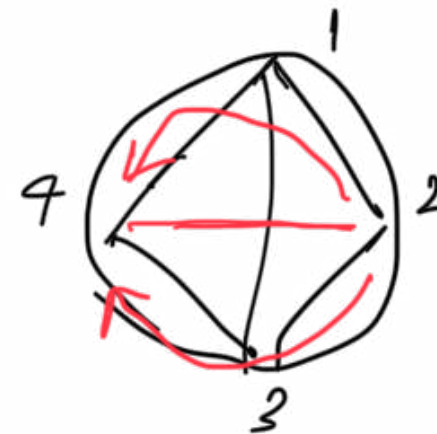
$$= \mathbf{Z}_2 + \lambda_3 \mathbf{Z}_3$$

$$(\mathbf{Z}_2 + \mathbf{Z}_3 - \mathbf{Z}_4) \lambda_1 - \mathbf{Z}_3 \lambda_3 = \mathbf{Z}_2$$

Note: $\infty \cdot (1,4) :$



명리서 보면 $\curvearrowright \approx \longleftrightarrow$
 $\rightarrow$
 근사 (approximation)



$(2,4)$ 에 대한 설명. By Kennedy theorem, $(2,4)$ and $\overrightarrow{(2,3) (3,4)}$ are $(2,1) (1,4)$

on a straight line (일직선 위)

HW 3-3, 12, 29, 62

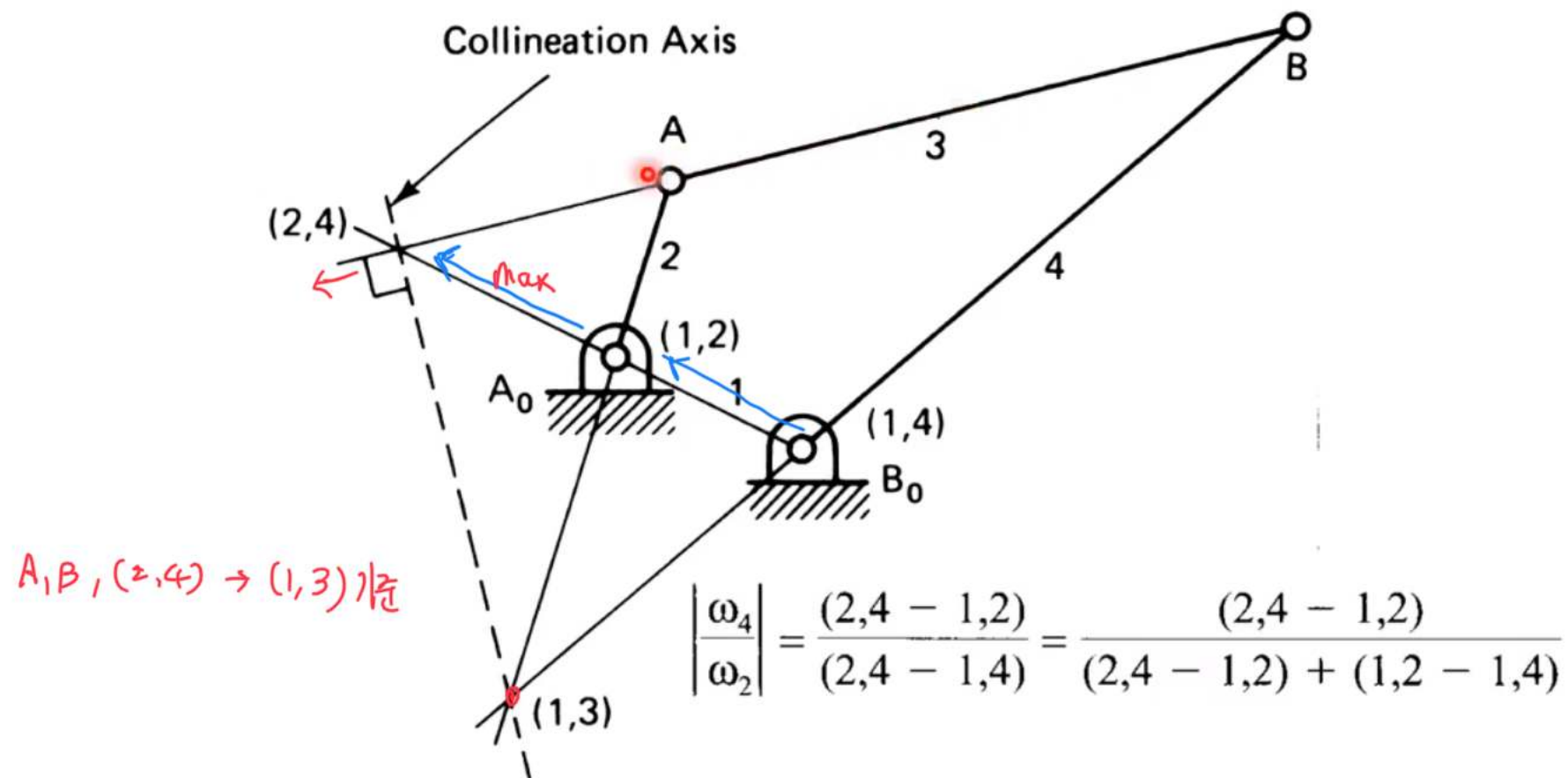


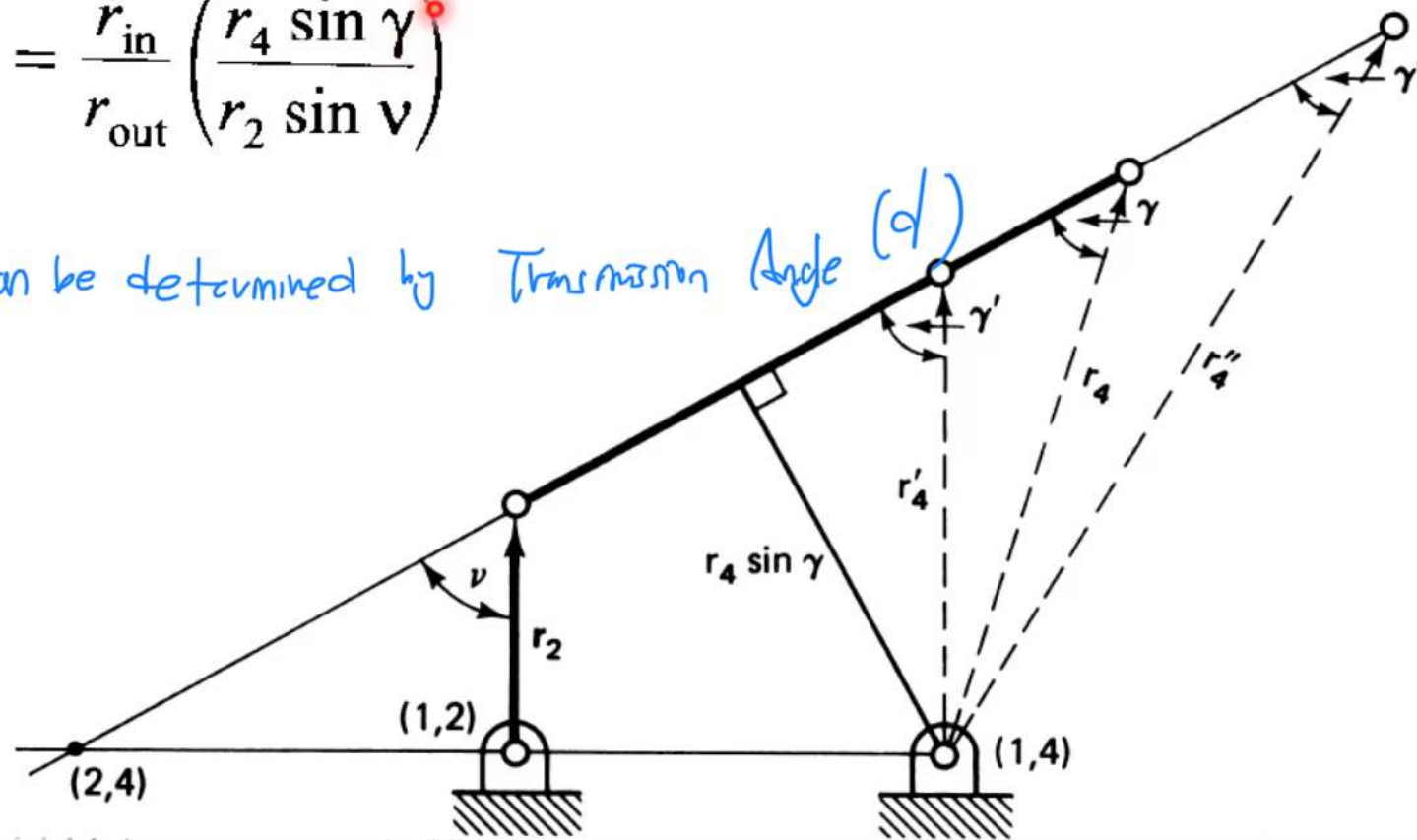
Figure 3.80 Freudenstein's theorem: At extreme of ω_4/ω_2 the collineation axis is perpendicular to coupler 3.

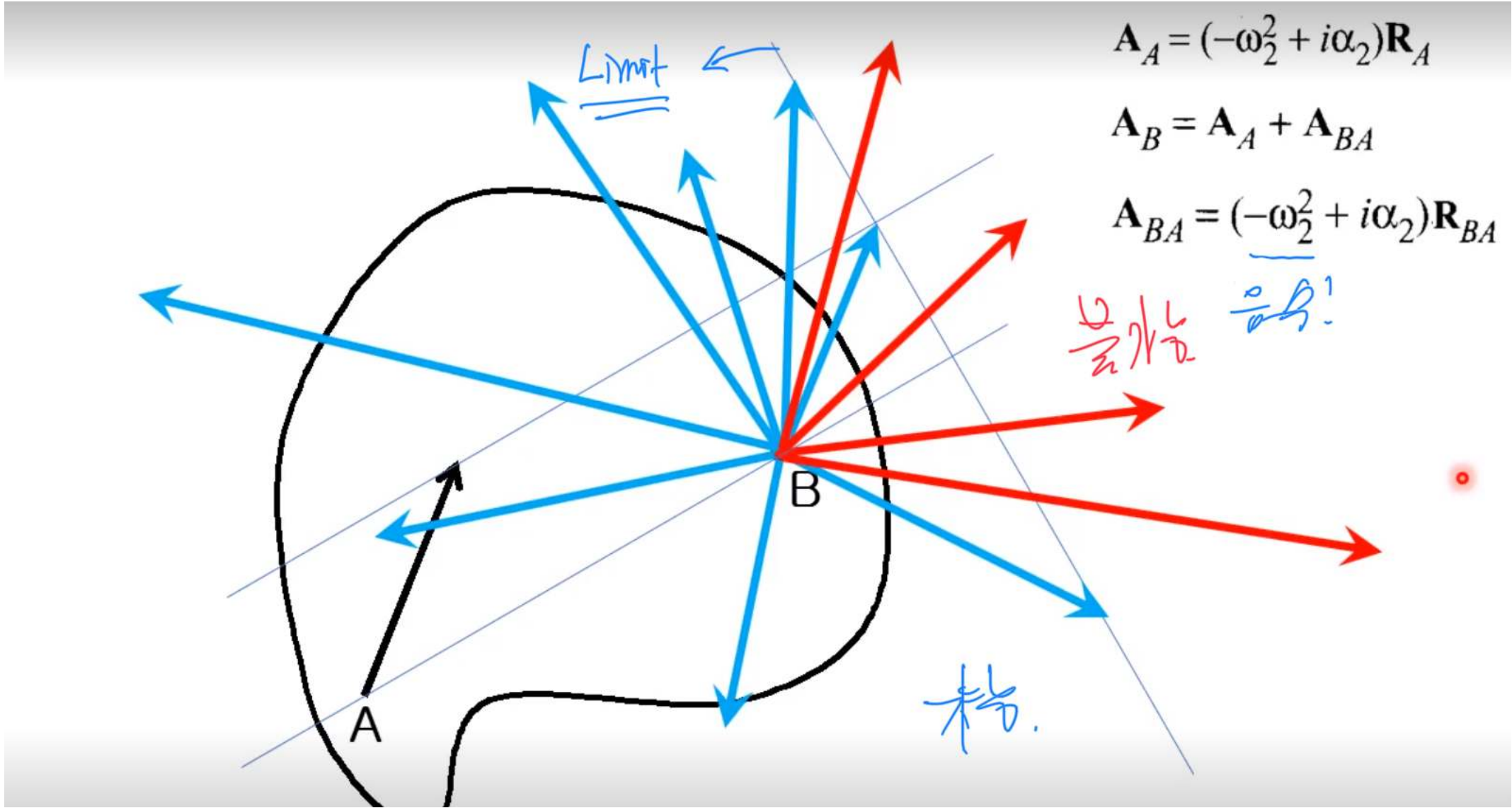
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$$\text{M.A.} = \frac{r_{\text{in}}}{r_{\text{out}}} \left(\frac{r_4 \sin \gamma}{r_2 \sin \nu} \right)$$

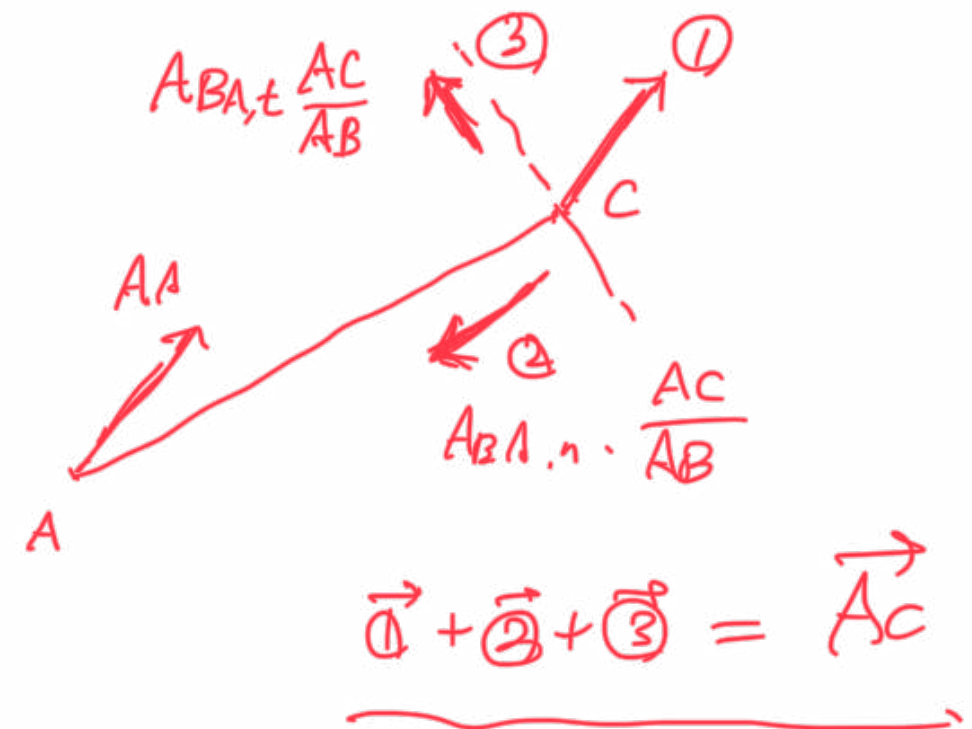
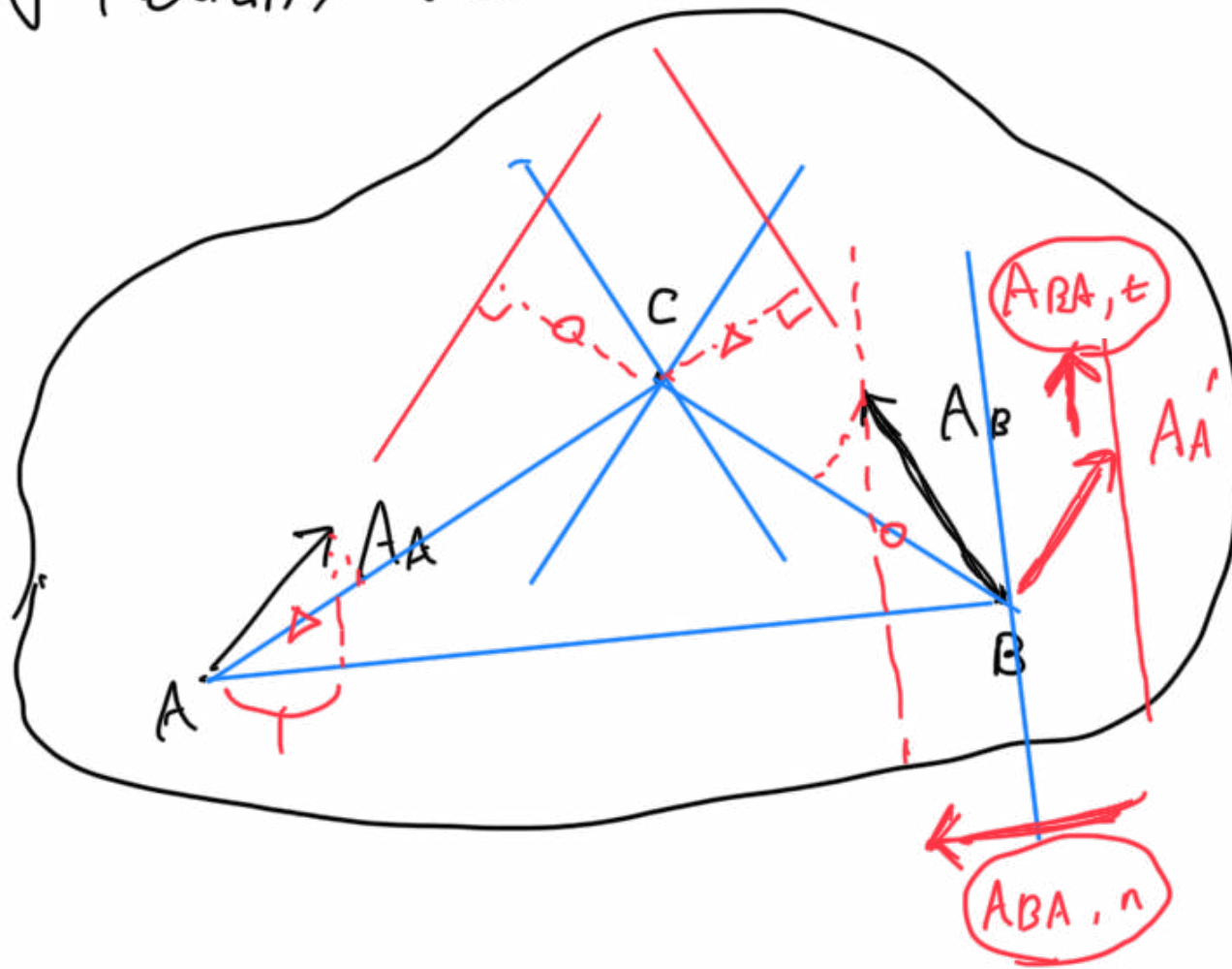
Can be determined by Transmission Angle (d)





가속도 제한 Acceleration Limitations

< My theorem > : Ac.



$\triangle ABC \sim \triangle abc$ (Goal)

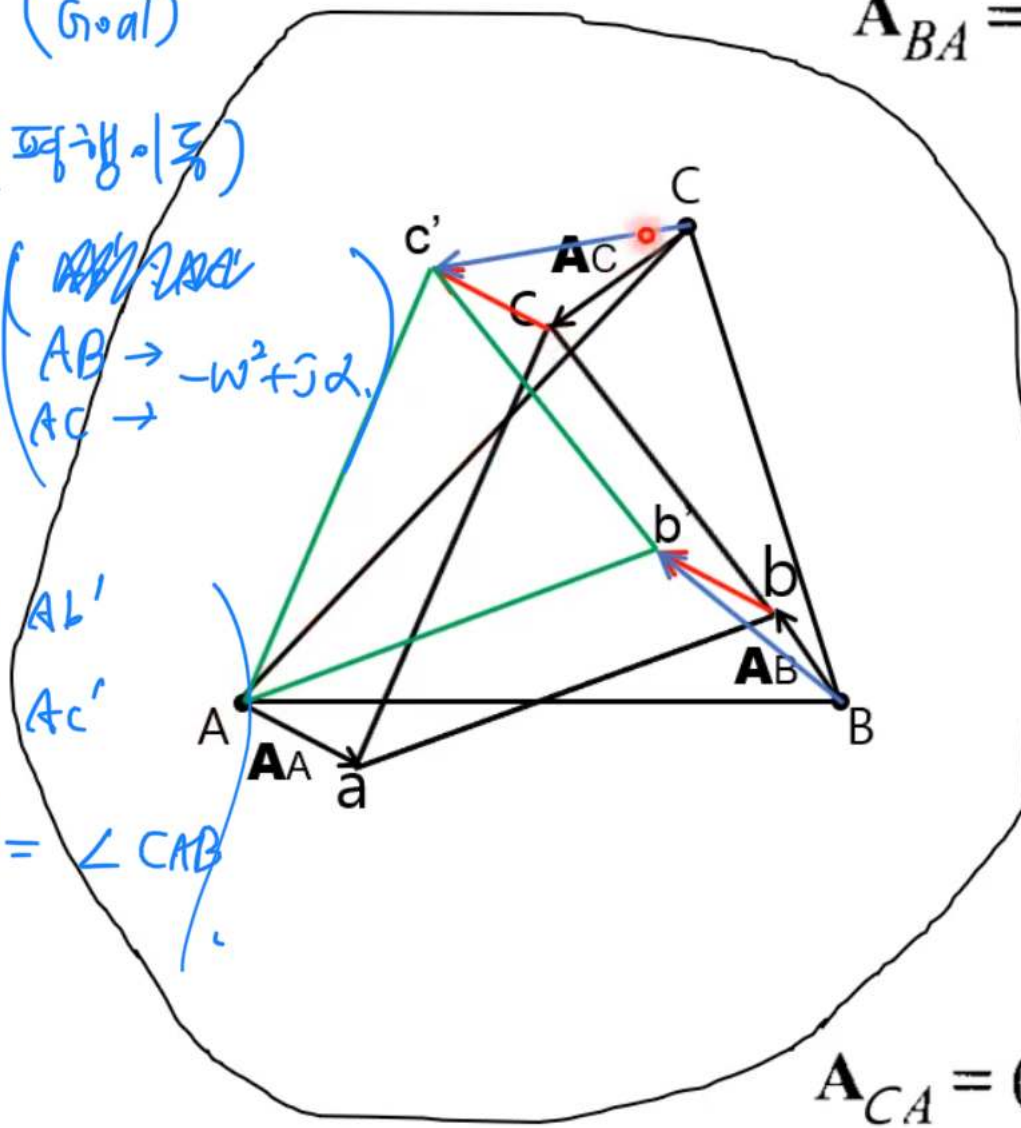
$\triangle Ab'c' = \triangle abc$ (평행이동)

$\triangle ABb' \sim \triangle ACc'$ (parallel)

$\triangle ABC \sim \triangle Ab'c'$ (AB $\rightarrow -\omega^2 + j\alpha$, AC $\rightarrow$)

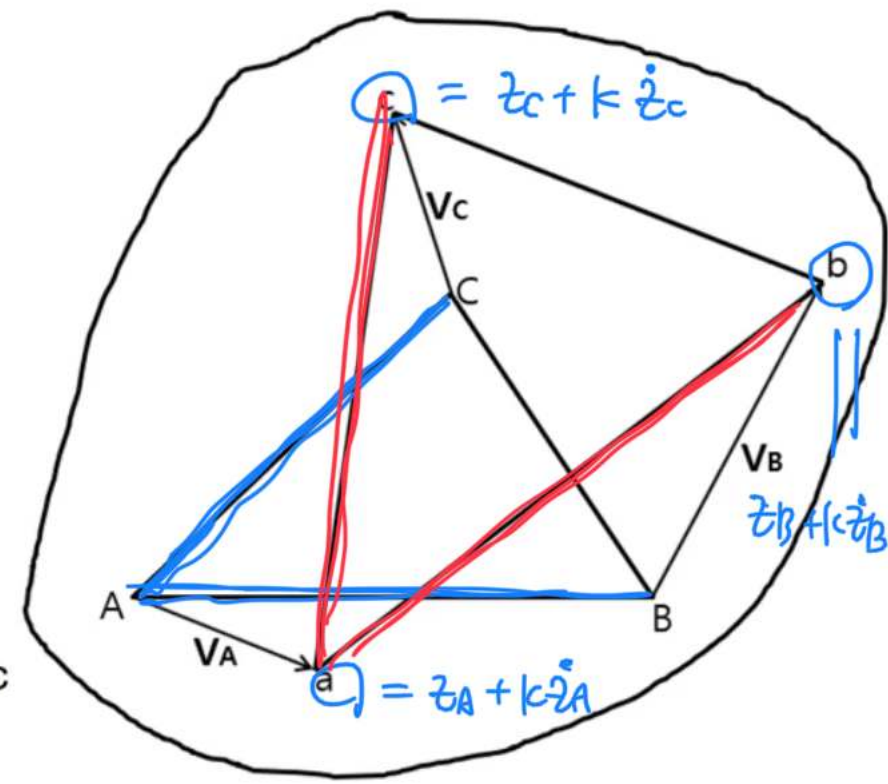
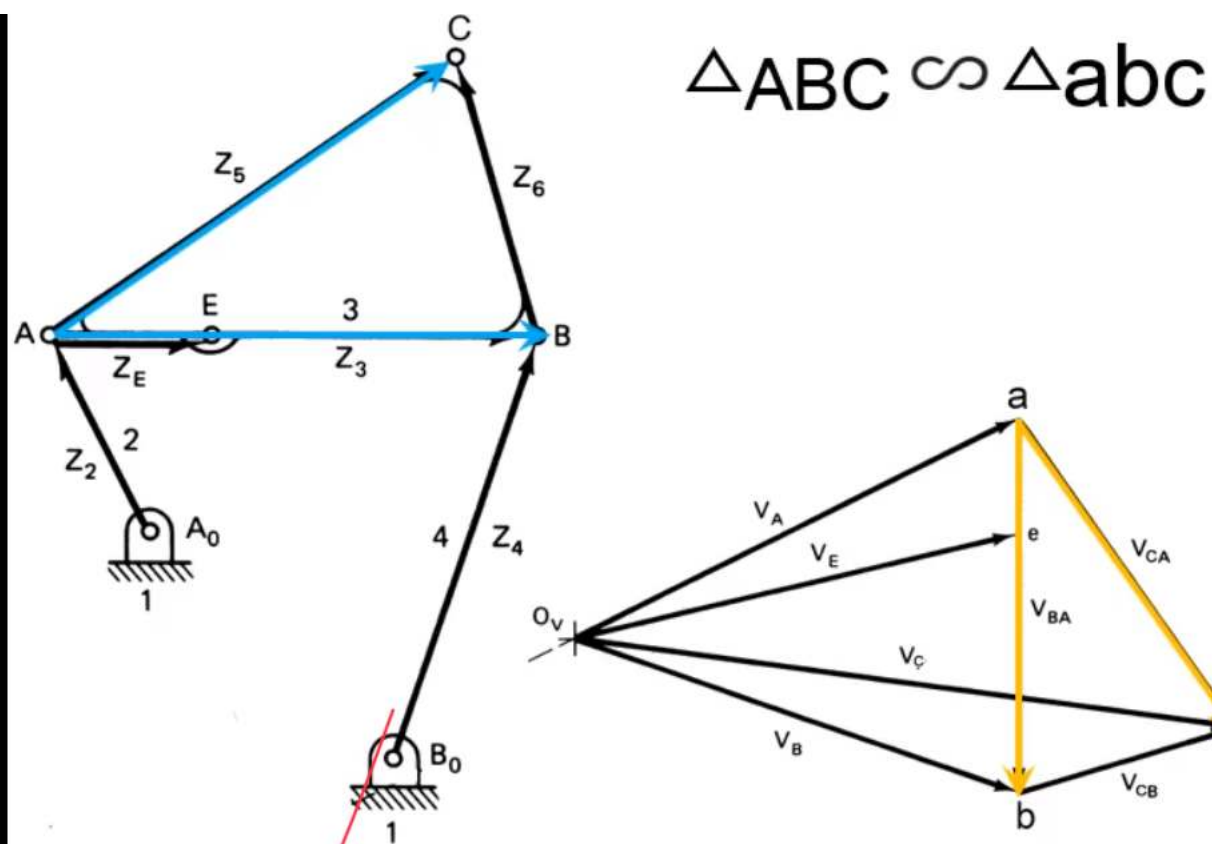
$\therefore AB:Ab'$
 $AC:Ac'$
 $\angle c'Ab' = \angle CAB$

$\Rightarrow \triangle ABC \sim \triangle abc$



$\mathbf{A}_{BA} = (-\omega_2^2 + i\alpha_2)\mathbf{R}_{BA}$

$\mathbf{A}_{CA} = (-\omega_j^2 + i\alpha_j)\mathbf{R}_{CA}$



$$Z_C = Z_A + C(Z_B - Z_A) \quad \dot{Z}_C = \dot{Z}_A + C(\dot{Z}_B - \dot{Z}_A)$$

$$C = \frac{Z_C - Z_A}{Z_B - Z_A} = \frac{\dot{Z}_C - \dot{Z}_A}{\dot{Z}_B - \dot{Z}_A} = \frac{Z_C - Z_A + k(\dot{Z}_C - \dot{Z}_A)}{Z_B - Z_A + k(\dot{Z}_B - \dot{Z}_A)} = \frac{Z_C + k\dot{Z}_C - (Z_A + k\dot{Z}_A)}{Z_B + k\dot{Z}_B - (Z_A + k\dot{Z}_A)}$$

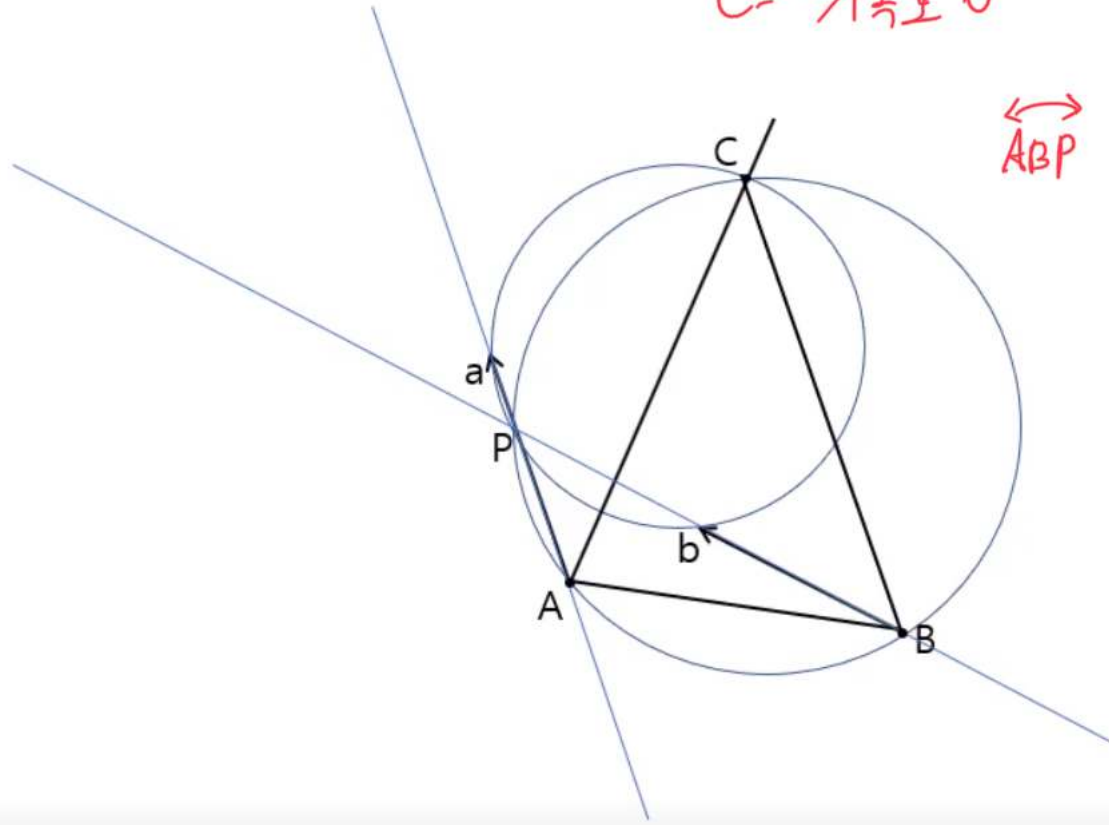
$$\Rightarrow \dot{z}_c = \dot{z}_A + C(\dot{z}_B - \dot{z}_A) \Rightarrow \ddot{z}_c = \ddot{z}_A + C(\ddot{z}_B - \ddot{z}_A)$$

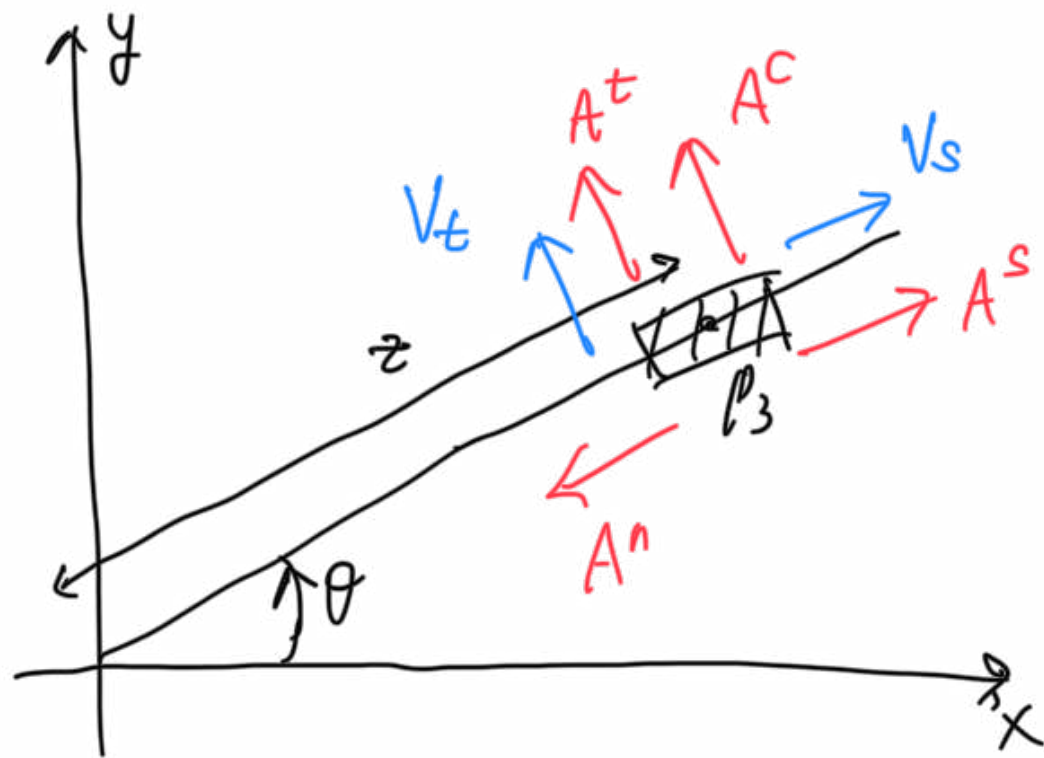
$$\Rightarrow \frac{z_C + k\dot{z}_C - (z_A + k\dot{z}_A)}{z_B + k\dot{z}_B - (z_A + k\dot{z}_A)}$$

same as velocity case.

$C = \text{가속도} \times t$

$\overleftrightarrow{ABP} \quad \overleftrightarrow{abp} \rightarrow C$





$$\begin{cases} \vec{z} = \underline{z} \underline{e^{j\theta}} \\ \text{scalars.} \end{cases} \quad V = dz/dt \quad \omega = d\theta/dt.$$

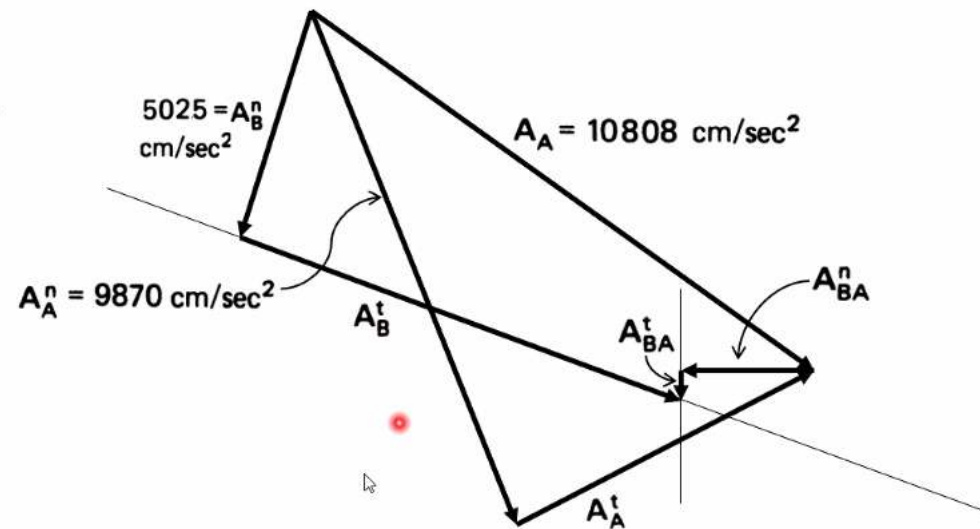
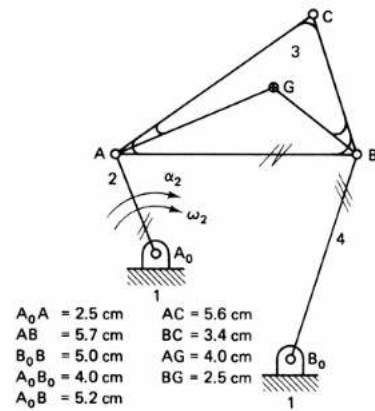
Position : $z e^{j\theta}$

Velocity : $\underline{v} e^{j\theta} + \underline{z} \omega j e^{j\theta}$

Acceleration : $\underbrace{a e^{j\theta}}_{\text{sliding}} + \underbrace{v \omega j e^{j\theta}}_{\text{Coriolis}} + \underbrace{v \omega j e^{j\theta}}_{\text{tangential}} + \underbrace{z \alpha j e^{j\theta} - z \omega^2 e^{j\theta}}_{\text{Centripetal (구심가속도)}}$

Centripetal : real pushing to center

Centrifugal : fake (pushes away from).



$$\mathbf{A}_A = \mathbf{A}_A^n + \mathbf{A}_A^t, \quad \text{where } |\mathbf{A}_A^n| = (A_0A)\omega_2^2, \quad |\mathbf{A}_A^t| = (A_0A)\alpha_2$$

$$\begin{array}{ccccccc} \mathbf{A}_B^n & + & \mathbf{A}_B^t & = & \mathbf{A}_A^n & + & \mathbf{A}_A^t & + & \mathbf{A}_{BA}^n & + & \mathbf{A}_{BA}^t \\ \text{D} \swarrow & & \text{D} \nwarrow & & \text{D} \swarrow & & \text{D} \nearrow & & \text{D} \nwarrow & & \text{D} \swarrow \\ \text{M} & & \text{M} & & \text{M} & & \text{M} & & \text{M} & & \text{M} \\ [(B_0B)\omega_4^2] & & [(A_0A)\omega_2^2] & & [(A_0A)\alpha_2] & & [(BA)\omega_3^2] & & & & \end{array}$$

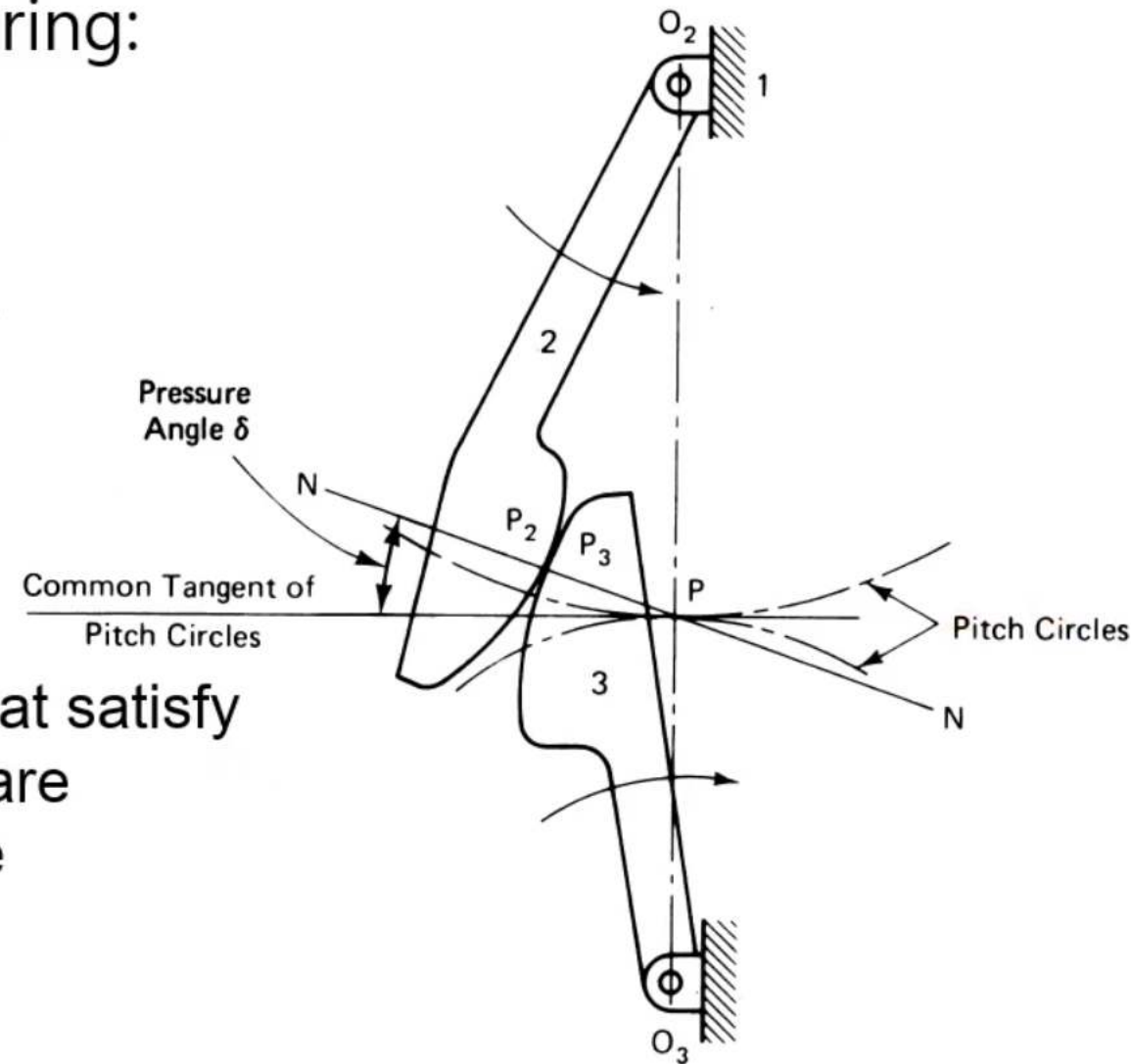
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The fundamental law of gearing:

For constant velocity ratio, the common normal of contacting tooth flanks must always pass through the pitch point P .

$$\left| \frac{\omega_2}{\omega_3} \right| = \frac{O_3P}{O_2P}$$

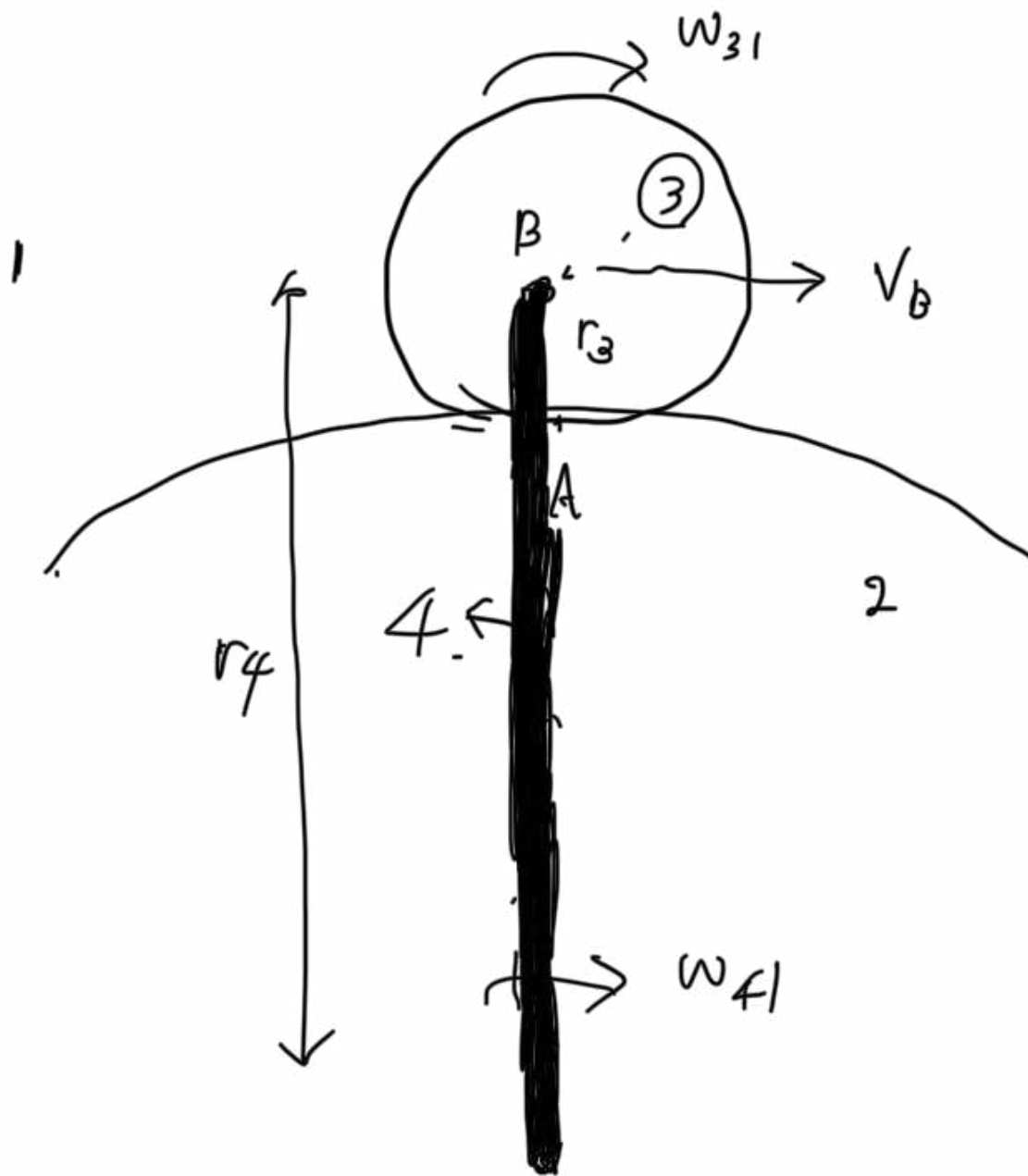


Any two mating tooth profiles that satisfy the fundamental law of gearing are called conjugate profiles and the condition is called conjugacy.

cycloidal profile

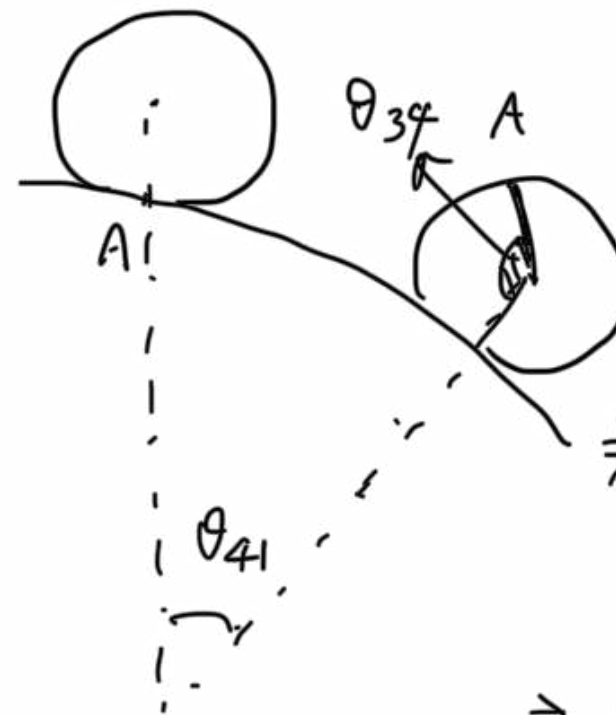
$P_2 P_3$ (perpendicular)

< Formula Method >



$$r_3 \omega_{31} = r_4 \omega_{41} = \|\vec{V}_B\|$$

$$\Rightarrow \omega_{31} = \frac{r_4}{r_3} \omega_{41} \quad (1)$$



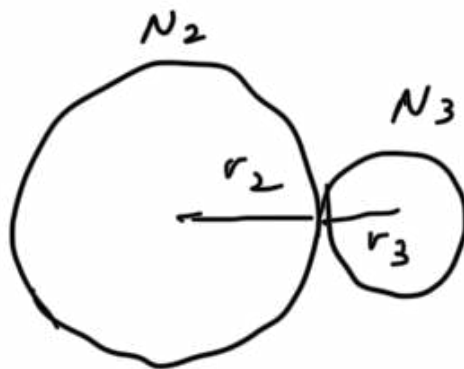
$$\theta_{34} \cdot r_3 = \theta_{41} (r_4 - r_3)$$

$$\Rightarrow \omega_{34} r_3 = \omega_{41} (r_4 - r_3)$$

$$\Rightarrow \omega_{34} = \left(\frac{r_4}{r_3} - 1 \right) \omega_{41}$$

$$\Rightarrow \omega_{34} = \left(\frac{r_2 + r_3}{r_3} - 1 \right) \omega_{41} \quad (2)$$

Since



$$\frac{N_3}{N_2} = \frac{r_3}{r_2} \Rightarrow \omega_{34} = \frac{N_2}{N_3} \omega_{41}$$

$$\omega_{34} = \frac{r_4}{r_3} \omega_{41} - \omega_{41} \Rightarrow \omega_{31} = \omega_{41} + \omega_{34}$$

or,

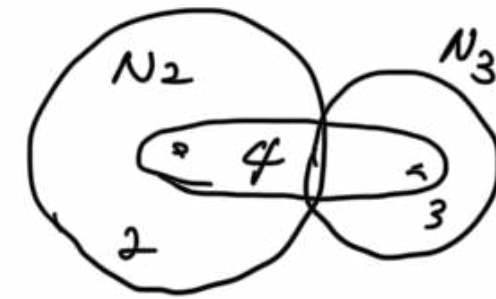
$$\omega_{31} = \omega_{41} + \omega_{34}$$

, Relative angular velocity

✓ relative angular velocity.
(Addition).

$$\omega_{21} = \omega_{41} + \omega_{24} \quad \text{Similarly,} \quad \omega_{34} = \omega_{31} - \omega_{41}$$

$$\omega_{24} = \omega_{21} - \omega_{41}$$



ω_{ij} : j th gear is fixed. E.g. $\frac{\omega_{34}}{\omega_{24}} = - \frac{N_2}{N_3}$

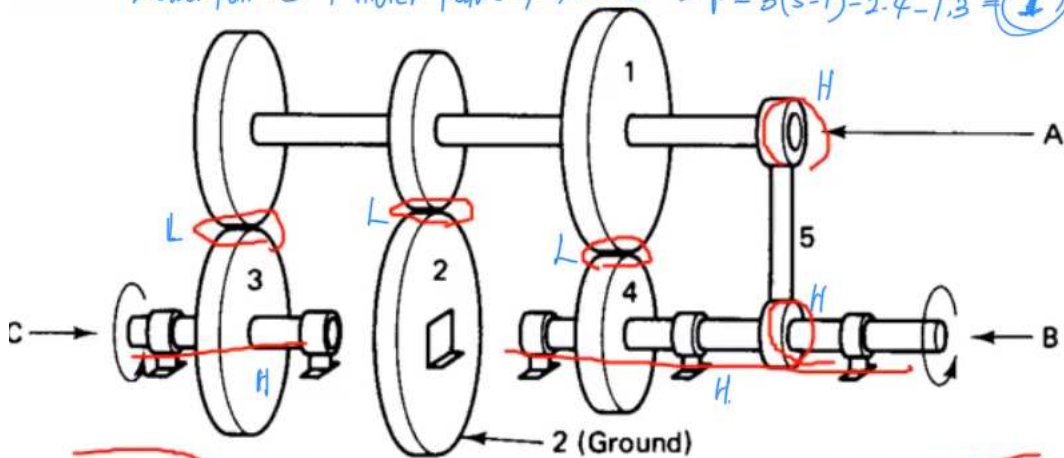
$$\Rightarrow \frac{|\omega_L|}{|\omega_F|} = \left| \frac{\omega_L - \omega_A}{\omega_F - \omega_A} \right| = \frac{\text{product of \# of teeth on driver gears.}}{\text{Product of \# of teeth on driven gears}}$$

부호 (Sign)

외접기어 (-) 내접기어 (+)

ω_A : Arm
 ω_L : Last gear
 ω_F : First gear

Lower Pair : 3, Higher Pair : 4, $h = 5 \Rightarrow F = 3(5-1) - 2 \cdot 4 - 1 \cdot 3 = 1$



$$\left| \frac{\omega_{LA}}{\omega_{FA}} \right| = \left| \frac{\omega_L - \omega_A}{\omega_F - \omega_A} \right| = \frac{\text{product of the numbers of teeth on driver gears}}{\text{product of the numbers of teeth on driven gears}}$$

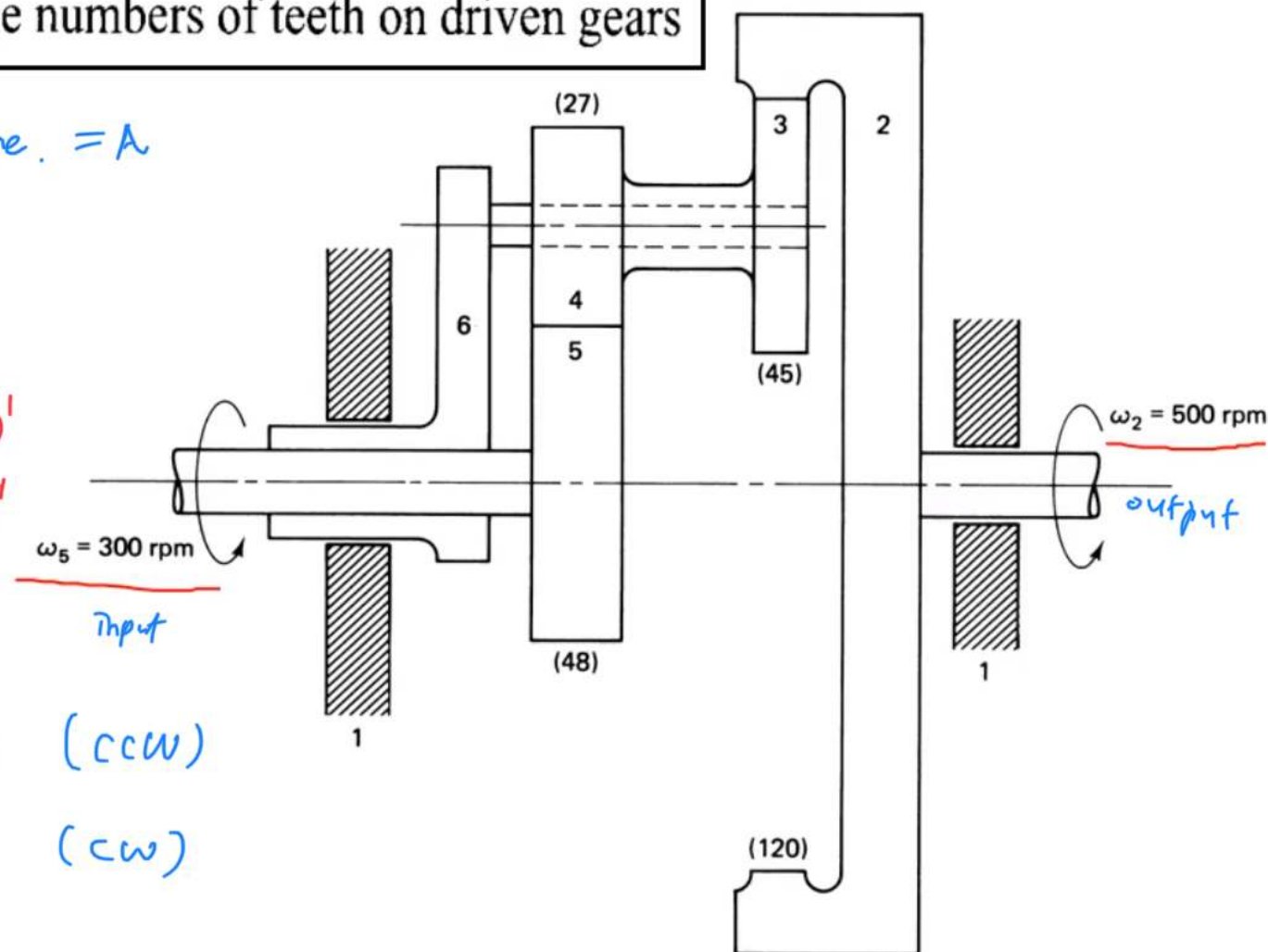
FORMULA METHOD

6 is the arm part of this machine. = A

$$\left| \frac{\omega_{26}}{\omega_{56}} \right| = \frac{\omega_2 - \omega_6}{\omega_5 - \omega_6} = - \frac{N_5 \cdot N_3}{N_4 \cdot N_2}$$

4,5 → 외접 (-)
2,3 → 내접 (+)

+ → Counter Clock Wise (ccw)
- → Clock Wise (cw)



FORMULA METHOD

$$\frac{\omega_{LA}}{\omega_{FA}} = \frac{\omega_L - \omega_A}{\omega_F - \omega_A} = \frac{\text{product of the numbers of teeth on driver gears}}{\text{product of the numbers of teeth on driven gears}}$$

$$\frac{\omega_{26}}{\omega_{56}} = \frac{\omega_2 - \omega_6}{\omega_5 - \omega_6} = \frac{N_3 N_5}{N_2 N_4}$$

$$= - \frac{(45)(48)}{(120)(27)} = -\frac{2}{3}$$

$$\frac{\omega_{26}}{\omega_{76}} = \frac{\omega_2 - \omega_6}{\omega_7 - \omega_6} = \frac{N_3 N_7}{N_2 N_4}$$

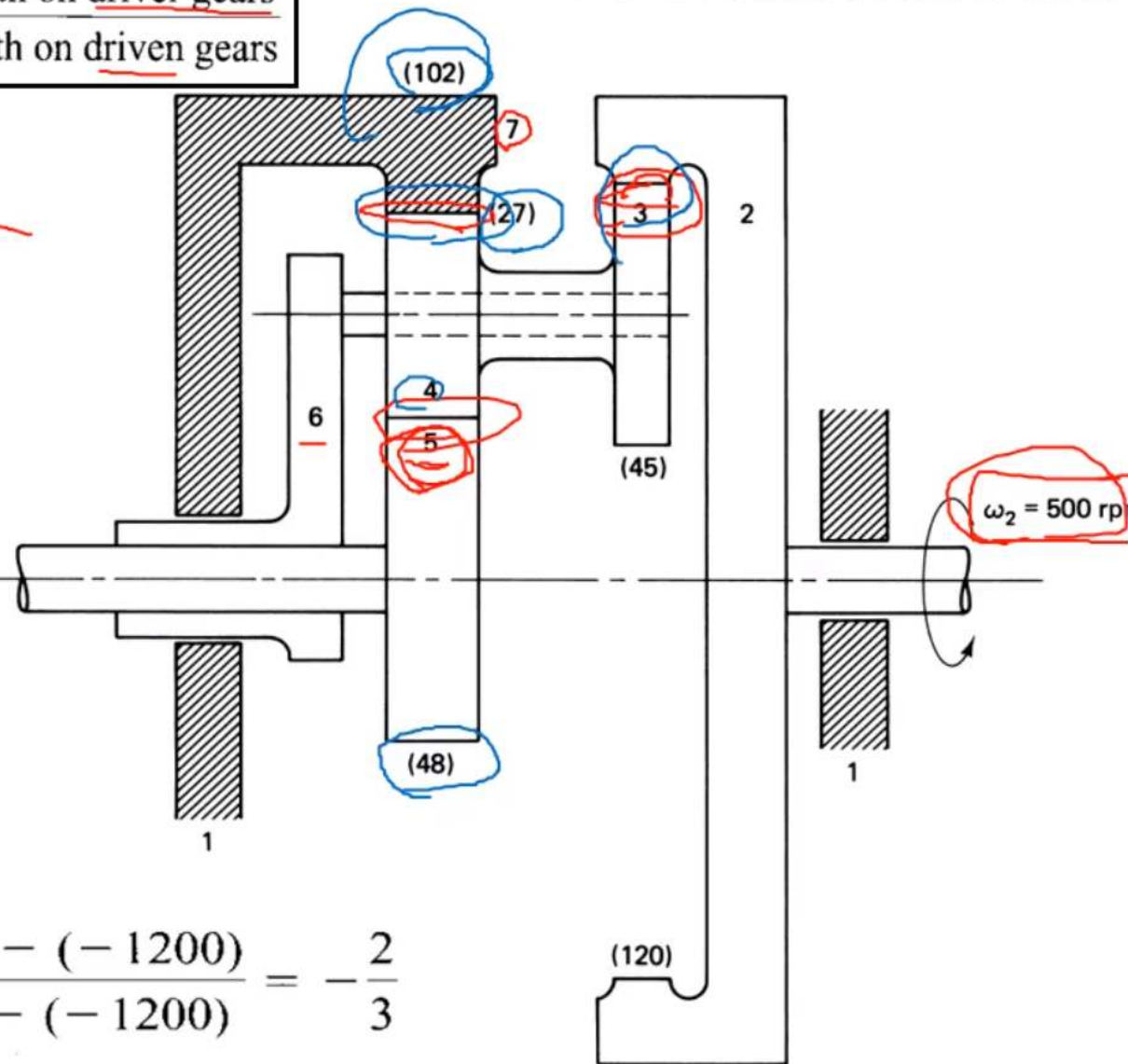
$$N_7 = N_5 + 2N_4 = 102$$

$$\frac{500 - \omega_6}{0 - \omega_6} = \frac{(45)(102)}{(120)(27)}$$

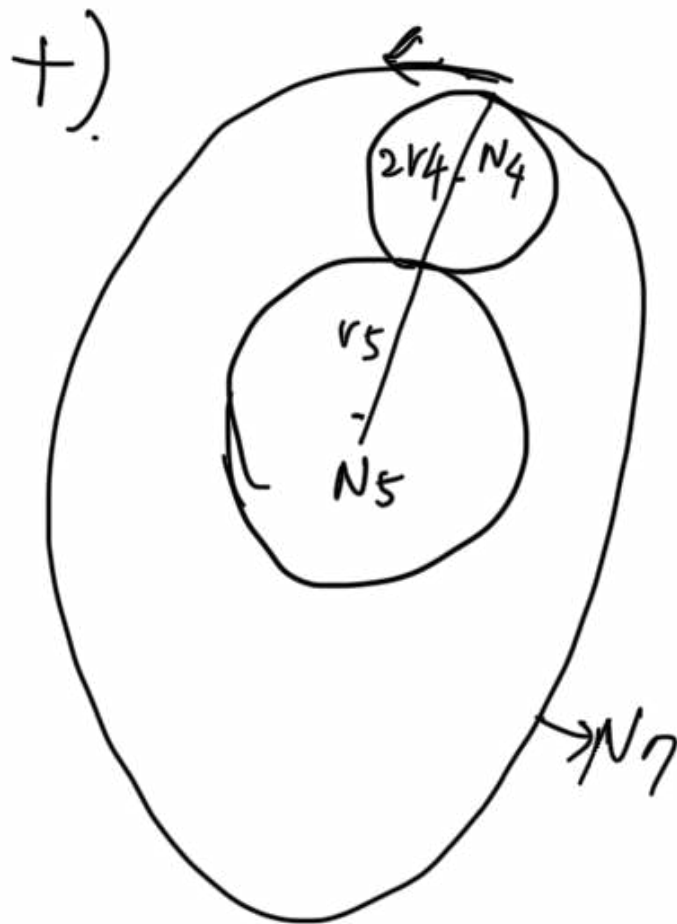
$$\omega_6 = \omega_6 \frac{(45)(102)}{(120)(27)} + 500$$

$$\omega_6 = -1200 \text{ rpm}$$

$$\frac{500 - (-1200)}{\omega_5 - (-1200)} = -\frac{2}{3}$$



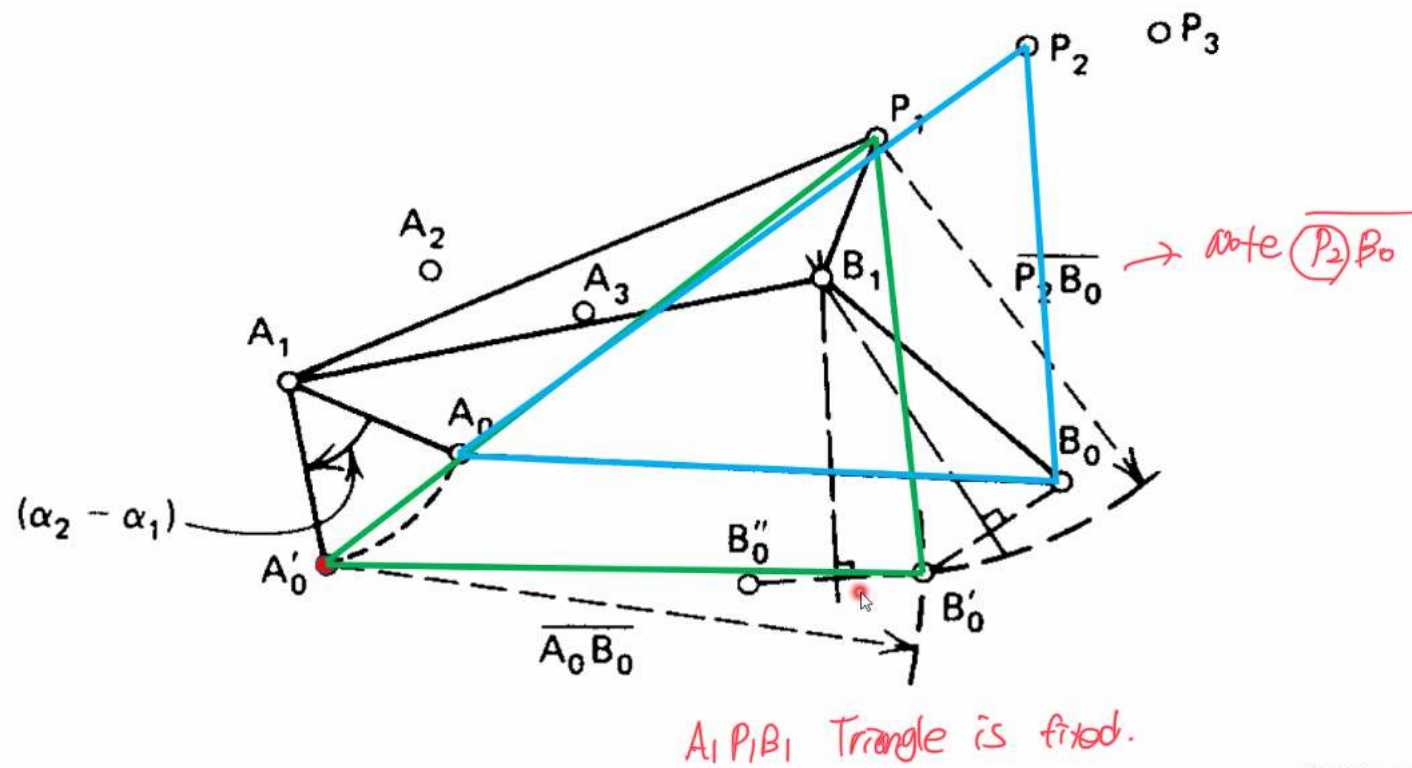
Why?



$$r_5 + 2r_4 = r_7$$

$$N_5 + 2N_4 = N_7 = 102$$

$$48 + 2 \cdot 27 \rightarrow$$



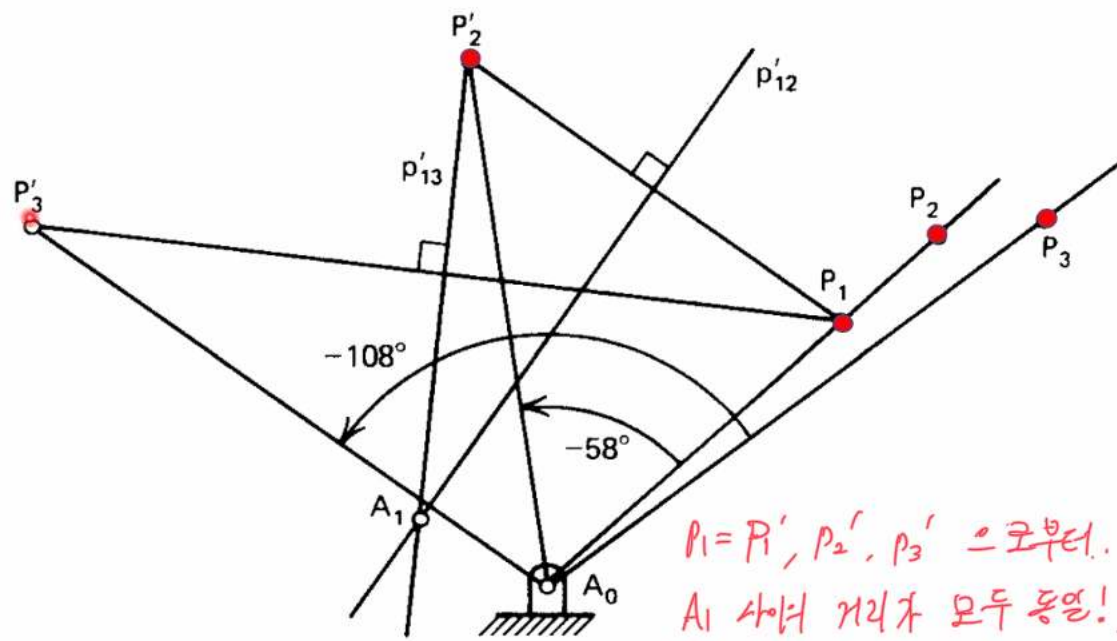
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Let



A hand-drawn diagram of a triangle with vertices labeled A_1 , A_2 , and A_3 . The interior angle at vertex A_1 is labeled 58° , and the interior angle at vertex A_2 is labeled 108° . Arrows point from the text labels to their respective angles.

is given and corresponding p_1, p_2, p_3 given.



$P_1 = P'_1, P_2 = P'_2, P_3 = P'_3$ 으로부터.
 A_1 사이의 거리가 모두 동일!
 ✕ kinematic Inversion!

PowerPoint 슬라이드 쇼 - [ch8-211130-실시간 zoom.pptx] - PowerPoint

GRAPHICAL SYNTHESIS FOR PATH GENERATION (WITHOUT PRESCRIBED TIMING): FOUR POSITIONS

point-position reduction method

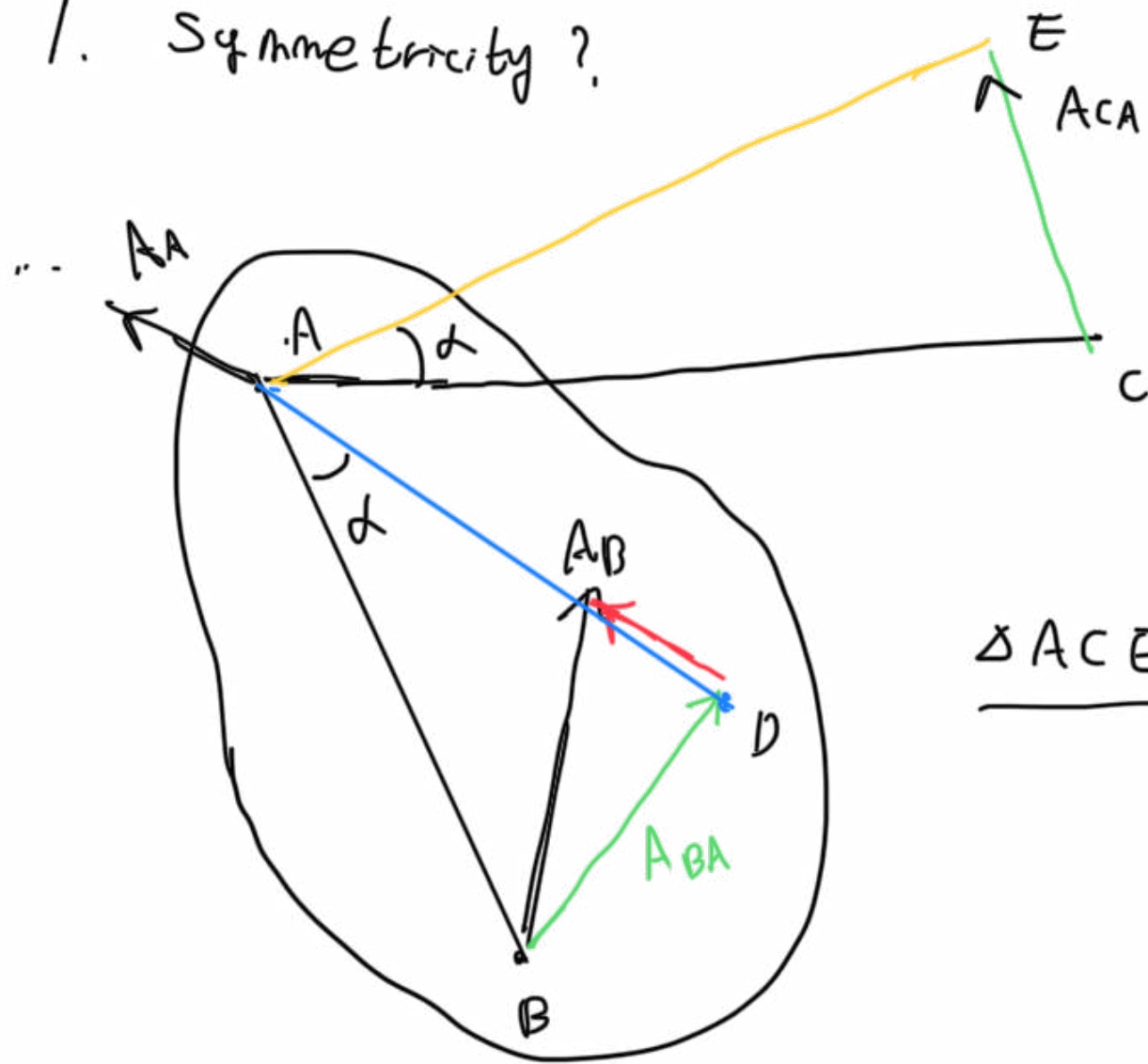
① P_1P_4 수직이등분선 $\rightarrow B_0$
 ② ψ_{14} 동일.

$\overline{A_1P_1} = \overline{A_4P_4}$
 $\triangle B_0A_4P_4 \equiv \triangle B_0A_1P_1$ (SAS)

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슬라이드 15/61

1. Symmetry?



$$\underline{\triangle ACE \sim \triangle ABD}$$

(ii) A_A, A_B, V_C 가 주어질 때 $\rightarrow V_D$

$$\vec{A}_B = \vec{A}_A + \vec{A}_{BA}$$

$$A_{BA} = (\underbrace{-\omega^2}_{\text{know}} + \underbrace{j\alpha}_{\text{know}}) \underbrace{\vec{R}_{BA}}_{\text{know}} \Rightarrow \omega, \alpha \text{ know.}$$

$$\vec{V}_D = \vec{V}_C + \vec{V}_{DC} = \underbrace{\vec{V}_C}_{\text{know}} + \underbrace{j\omega}_{\text{know}} \underbrace{\vec{R}_{DC}}_{\text{know}} \rightarrow \underline{\text{fixed}} V_D.$$

(ii) $V_A, V_B, \textcircled{A_C}$ 가 주어질 때. $\rightarrow A_D$

$$\vec{V}_B = \vec{V}_A + \vec{V}_{BA}$$

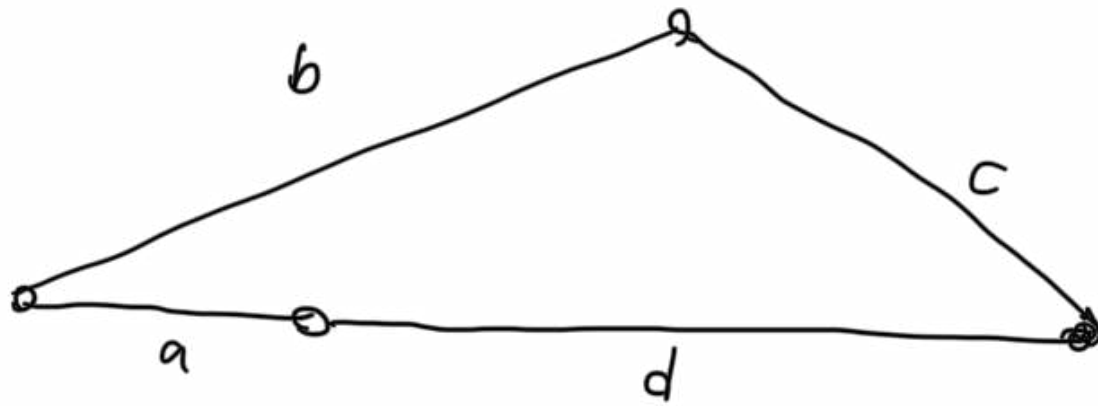
$$\vec{V}_{BA} = \underbrace{j(\omega)}_{\text{know}} \underbrace{\vec{R}_{BA}}_{\text{know}} \Rightarrow \omega \text{ know}$$

$$\vec{A}_D = \underbrace{\vec{A}_C}_{\text{know}} + \vec{A}_{DC} \rightarrow \text{unfixed } A_D$$

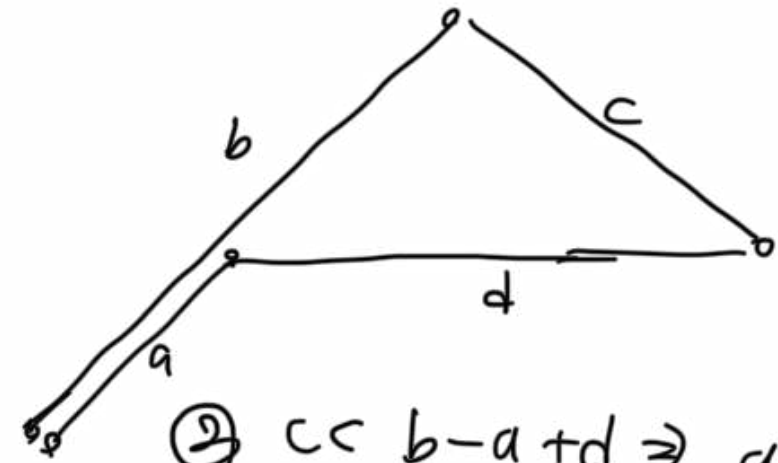
$$\underbrace{(-\omega^2 + j\alpha)}_{\text{know}} \underbrace{\vec{R}_{DC}}_{\text{unknown}}$$

$$b - a < c + d$$

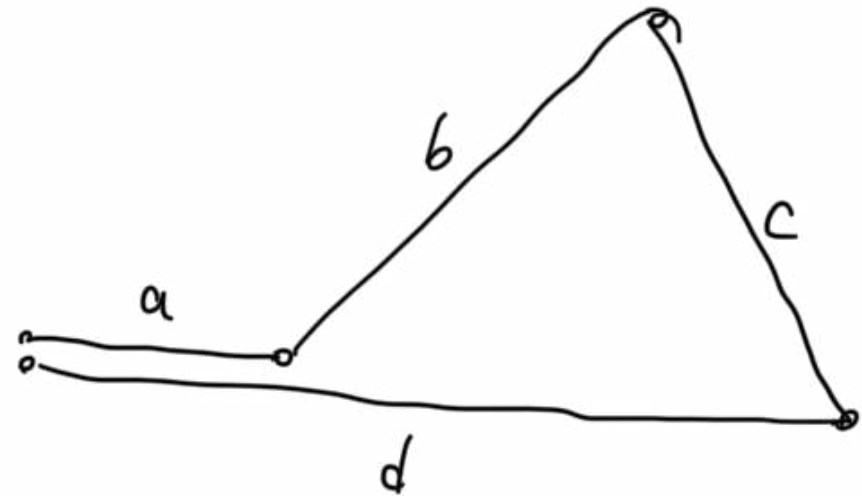
2. Grashof mechanism. $\begin{cases} a = s \text{ (shortest)} \\ d = l \text{ (longest)} \end{cases}$



$$\textcircled{1} a + d < b + c$$



$$\textcircled{2} c < b - a + d \Rightarrow a + c < b + d$$

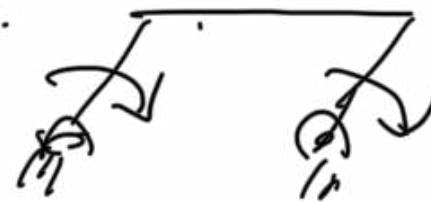


$$\textcircled{3} b < d - a + c \Rightarrow a + b < c + d$$

회전운동 ≠ 평직운동. but 회전운동하는 평직운동 존재.

$$\textcircled{1} a + \underline{d} < b + c$$

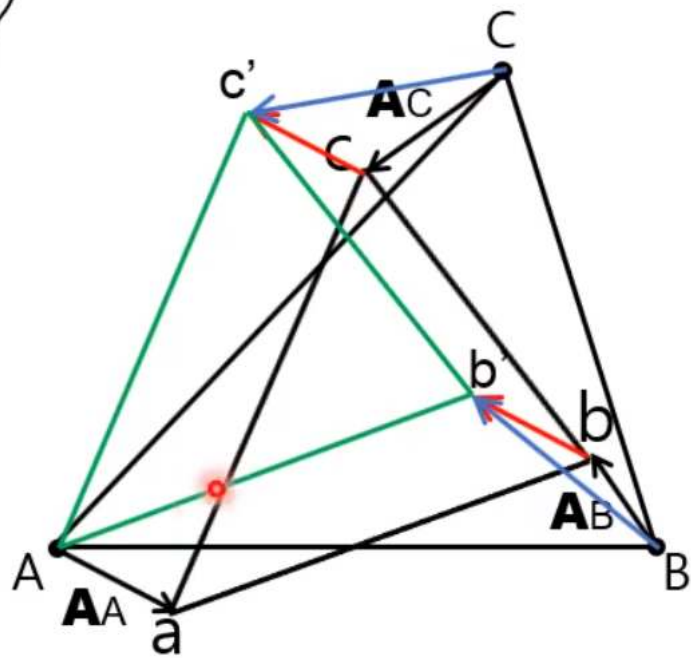
$$\textcircled{2} a + \underline{c} < b + d \Rightarrow s + l < p + q$$

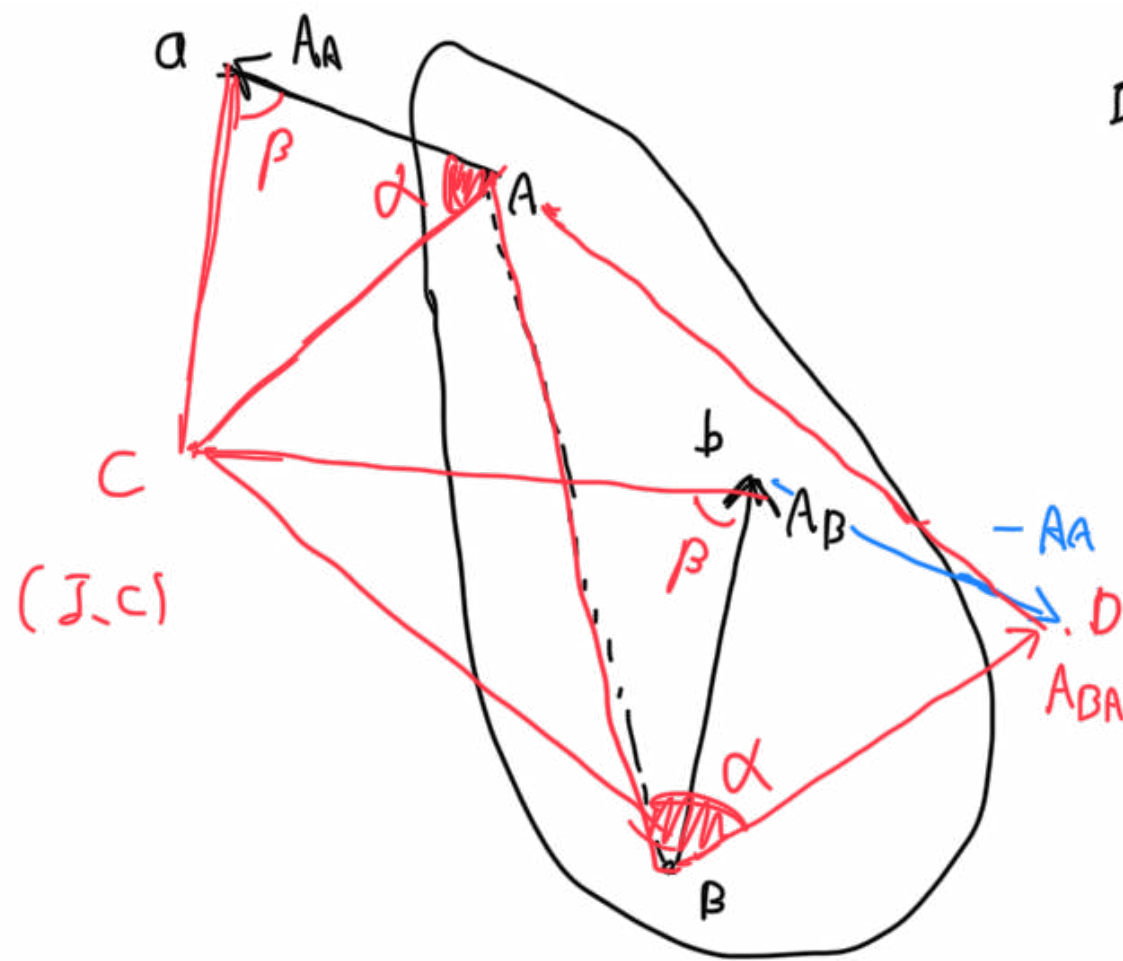


$$\textcircled{3} \quad a + \underline{b} < c + d$$

$\overline{a}$

3. I.C (Acceleration = 0)





$$\text{If } \vec{A}_C = \vec{0}, \quad \vec{A}_{CA} + \vec{A}_A = \vec{A}_C$$

$$\Rightarrow \vec{A}_{CA} = -\vec{A}_A$$

$$\underline{\vec{A}_{AC} = \vec{A}_A}$$

$$\vec{A}_{BA} = (-\omega^2 + j\alpha) \vec{R}_{BA}$$

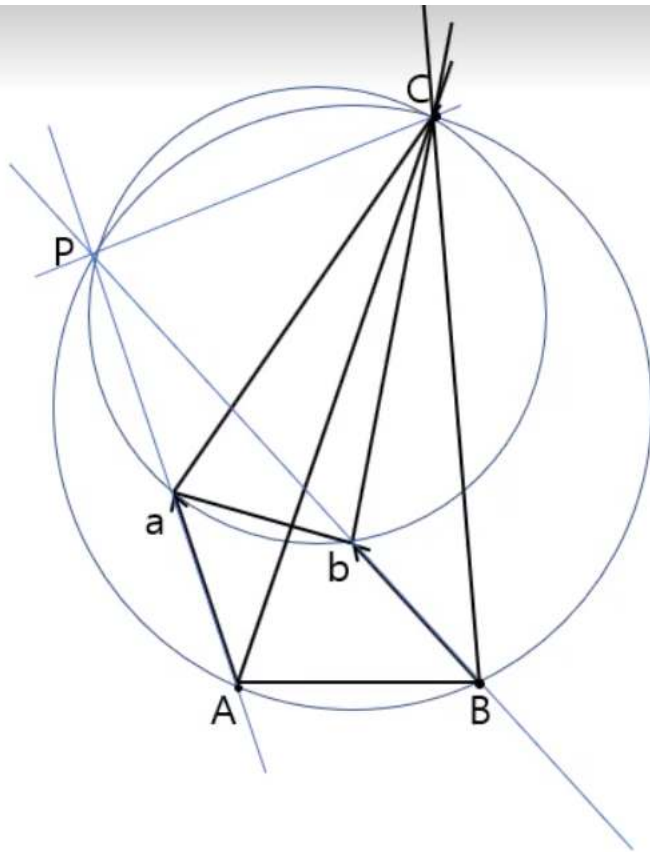
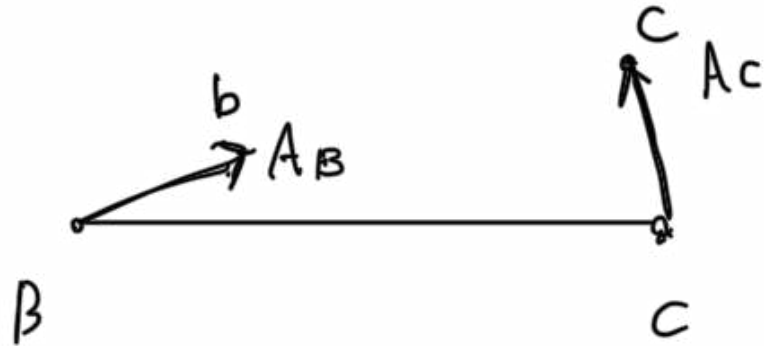
$$\vec{A}_{AC} = (-\omega^2 + j\alpha) \vec{R}_{AC}$$

Note : $\Delta Cab \sim \Delta CAB$

if $A_A = 0$, (A is I.C of acceleration), it's as follows

There exists A , s.t. $ABC \sim Abc$

$\Rightarrow$ Find A $\rightarrow$ I.C. of acceleration.



$$\vec{r}_p = r e^{j\theta_2} \quad \dot{\vec{r}}_p = \dot{r} e^{j\theta_2} + r \omega_2 j e^{j\theta_2}$$

$$\dot{r} \cos \theta_2 - r \omega_2 \sin \theta = 0$$

$$\underline{\dot{r} = r \omega_2 \tan \theta = 32}$$

$$\vec{v}_p = 32 e^{j\theta_2} + \left(24 \cdot \frac{2}{\sqrt{3}}\right)_{1,2} \cdot j e^{j\theta_2} = \textcircled{64} \uparrow$$

$$\vec{a}_p = \ddot{r} e^{j\theta_2} - r \omega_2^2 e^{j\theta_2} + 2 \dot{r} \omega j e^{j\theta_2}$$

$$(\ddot{r} - r \omega_2^2) \cos \theta - 2 \dot{r} \omega \sin \theta = 0$$

$$\ddot{r} - r \omega_2^2 = 2 \dot{r} \omega \tan \theta$$

$$\Rightarrow \ddot{r} = \textcircled{\frac{320}{\sqrt{3}}}$$

73.9 → + ↖ 128